India Semiconductor Industry

Market Opportunity and Strategic Investment Analysis, 2026–2030

Case studies: Micron Technology (Sanand) and Tata Electronics (Dholera and Assam)

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Portfolio case study. All figures in ₹ are converted at ₹95 per US$ unless a company disclosed the US$ figure itself. Reference facilities are illustrative and are not company economics.

Contents

Executive summary 1
1. Introduction and approach 2
2. The global semiconductor industry 2
3. The Indian semiconductor market 3
4. Government policy and incentives 6
5. How India compares with other chip hubs 7
6. The wider Indian ecosystem 9
7. Case study: Micron's Sanand assembly and test plant 10
8. Case study: Tata Electronics – Dholera fab and Assam OSAT 13
9. Financial analysis 15
10. Strategic recommendations 18
11. Conclusion: testing the hypothesis 20

References 21
Annex A. Methodology 23
Annex B. Industry, policy and ecosystem detail 25
Annex C. Company case detail 27
Annex D. Key assumptions 31
Annex E. Data cautions and limitations 31
Annex F. Glossary 32
Annex G. Notebooks and workbook 33

Executive summary

Question. Can India become a globally competitive chip manufacturing and supply-chain hub by 2030, and which parts of the value chain are most attractive? **Answer: partly.** By 2030 India will be a credible new entrant in chip packaging, with about 1–2% of global assembly and test capacity. Its plants will be cost-competitive only because of subsidies. A mature-node fab is possible but will lose money until the early 2030s. Leading-edge fabrication is not realistic.

Situation. India used about US$52 billion of chips in FY2024-25, roughly 6–8% of world demand, and imported almost all of them. Since 2021 the government has committed about ₹2 lakh crore through Semicon 1.0 and 2.0. Twelve projects worth ₹1.64 lakh crore are approved, and three packaging plants (Micron, Kaynes and CG Semi) started commercial production in 2026.

Complication. Most of this capacity packages wafers made abroad. In the base case for 2030, plants in India make or package 7.1% of what India uses, but only 1.7% reaches Indian buyers and only 1.3% of the value is added in India. The economics depend on the subsidy. My reference packaging plant earns a 10-year IRR of −1.1% without support and 15.9% with Semicon 1.0-level support. A Dholera-sized fab needs more than 100% loading to break even without support. Semicon 2.0 has now cut support rates.

Resolution. India should treat packaging as its main manufacturing bet and use the subsidy to win customers, not just to build floor space. It should finish Dholera but not fund a second fab on Semicon 2.0's current terms. The bigger prize is turning India's design talent into Indian-owned chip IP.

Five numbers that carry the report

Measure	Value	Section
India chip consumption, FY25 → 2030 (base)	US$52B → US$96B	3
India 2030: made or packaged / supplied to Indian buyers / value added	7.1% / 1.7% / 1.3% of demand	9.1
India's share of global assembly and test, 2030 (base, by value)	~1.6% (Malaysia ~7% today)	5.4
Reference packaging plant, 10-year IRR: no support vs ~65% support	−1.1% vs 15.9%	9.2
Reference fab, break-even loading in 2033: no support vs ~65%	128% vs 60% (20-yr IRR at best 7.0%)	9.3

Top recommendations

1. **Make packaging the core, and pay for customers, not only capex.** Utilisation and price move a plant's IRR far more than labour or yield. Track: utilisation of merchant packaging plants.
2. **Finish Dholera; do not fund a second fab on Semicon 2.0 terms.** On eligible-capex rules a new fab's 20-year IRR is about 1.8%. Track: Dholera's loading in 2030–32.
3. **Turn design talent into owned IP.** India has ~20% of the world's chip designers but Indian design firms earn under ₹150 crore. Track: revenue of Indian-owned design companies.
4. **Build the supplier base.** About 55% of a packaging plant's cost is materials, almost all imported. Track: the local share of materials in Indian packaging plants.

1. Introduction and approach

India wants to make chips, not just use them. This report asks whether that is realistic by 2030, and where in the chain India should focus. It is written for investors and policy readers, and every number can be traced to a source or to the Excel workbook that comes with it.

Starting hypothesis. By 2030, India has a realistic opportunity to become a competitive player in assembly, testing and packaging, and in selected mature-node and compound-semiconductor manufacturing. Leading-edge fabrication will remain difficult because of technology, capital, ecosystem and supply-chain constraints. I test this hypothesis rather than assume it, and Section 11 gives the verdict.

What "globally competitive" means here. I use three measurable criteria:

1. **Domestic manufacturing contribution.** How much of India's chip demand is met by plants in India, separating chips fabricated in India from imported wafers packaged here, and how much value is actually added in India.
2. **Position in the global supply chain.** India's share of global packaging and fab capacity, and growth in design, packaging and fab capability.
3. **Manufacturing economics.** Whether an Indian plant can match Malaysia and Vietnam on cost, looking at capex, labour, power, incentives and suppliers.

Approach. The work covers the global industry, the Indian market, policy, benchmarking and the wider ecosystem, the two company cases and a financial model. Sources are official first, then company filings, then industry bodies and reputable research (Annex A.1). Every calculation is reproducible in the workbook or one of five notebooks (Annex G); data cautions are in Annex E.

Conventions used throughout. These apply to every section, so I state them once here:

- **Exchange rate.** One rate, ₹95 per US$. Figures a company disclosed in US$ (Micron US$2.75 billion, CG Semi's US$500 million target, UMC US$5 billion) stay in US$.
- **Captive and unlisted companies.** Micron's Sanand plant is captive, so I measure the value of products passing through it ("throughput value"), not its revenue or profit. Tata Electronics is unlisted, so Dholera is valued only at peer benchmark prices ("illustrative output value"), not as a Tata revenue forecast.
- **Reference facilities.** Where I need profit and return figures, I use a reference packaging plant (built from Amkor) and a reference fab (built from UMC). These are illustrative and are **not** the economics of Micron, Tata, CG or any other company.
- **Support rates.** Semicon 1.0 and 2.0 rates are summarised once, in Exhibit 4 (Section 4.2).

2. The global semiconductor industry

> **Key takeaways**
>
> - Normal market growth is about 7–9% a year. 2026 is an abnormal memory-boom year, and I keep it out of the trend.
> - Design and leading-edge fabs hold most of the value. Packaging is only about 6% of value added, but it is the easiest manufacturing step to enter.
> - Fab, packaging and equipment capacity is concentrated in China, Taiwan and Korea.

2.1 How the value chain works

A chip goes through four stages: design, wafer fabrication ("front end"), assembly, test and packaging (ATP, or "back end"), and the supply of equipment and materials. **Fabless** companies design chips (Qualcomm, Nvidia); **foundries** make chips designed by others (TSMC, UMC, PSMC); **IDMs** design and make their own (Intel, Samsung, Micron); and **OSATs** package and test chips for others (ASE, Amkor), called ATMP in Indian policy documents (Annex F). **Logic** chips come in many designs and are often made at foundries. **Memory** (DRAM and NAND) is a near-commodity made by three or four IDMs, so its price swings much more.

2.2 Market size and growth

The world chip market was US$335.2 billion in 2015 and US$795.6 billion in 2025.[1] That is 9.0% a year, or 7.3% a year if I stop at 2024, before the boom. WSTS's Spring 2026 forecast puts 2026 at US$1,511 billion (+90%), driven by memory (+250%). Its Q2 update, using actual sales to June, implies about US$1,655 billion. Either figure is far above the trend of about US$854–867 billion, so I treat 2026 as a one-off. WSTS's Autumn 2026 forecast has not yet been released, so global figures in this report use the Spring 2026 forecast plus the Q2 update.

2.3 Where value and capacity sit

Exhibit 1: Design and fabrication capture three-quarters of the industry's value; packaging captures about 6%

Activity	Share of value added, 2022	Who leads (by capacity or HQ)
Design (logic, memory, analog/discrete)	56%	US-headquartered firms
EDA tools and IP	3%	US
Equipment	12%	US, Japan, Netherlands
Materials	5%	Japan, Korea, Taiwan
Wafer fabrication	19%	China 24%, Taiwan 18%, Korea 17%, Japan 17% of capacity
Assembly, test and packaging	6%	China 30%, Taiwan 27%, Korea 9%; Malaysia ~7%

Source: SIA–BCG (2024), Exhibits 2, 6 and 13 (Ref. 24). Value-added shares by company headquarters; fab and ATP shares by plant location.

The concentration is sharpest at the leading edge. Taiwan held 69% and Korea 31% of capacity below 10 nm in 2022. SIA–BCG expects Vietnam and Malaysia to gain packaging share by 2032 (to about 8% and 9%). India does not appear separately in any of these datasets.

In simple terms: a fabless firm sells its ideas, a foundry sells factory time, and an OSAT sells a low-margin service on expensive machines (17% gross margin and 16% capex intensity in 2016–19; Annex B.1, Exhibit A5). Amkor's cost of sales is 55% materials, 10% labour and 9% depreciation, so materials, not labour, are the largest cost. Memory is the most cyclical model: Micron's gross margin went from −9% in FY2023 to 81% in FY2026.

What this means. Packaging is the most accessible manufacturing entry point for a new country. It needs less capital and know-how than a fab. But it captures little value, and its margins are thin. Design is where most value sits, and India already has the people for it (about 20% of the world's chip-design engineers). Leading-edge fabs, memory fabs and equipment are out of reach for India by 2030 on capital and know-how alone.

3. The Indian semiconductor market

Key takeaways

- India used about US$52 billion of chips in FY2024-25 (IESA). In my base case this reaches about US$96 billion by 2030.
- Loose-chip imports cover only about half of that. The rest arrives inside imported boards and modules, so real import dependence is close to 100%.
- Handsets are more than half of demand. Plants approved so far would earn only 1.5–3.1% of the 2030 market (base 2.3%).

3.1 What we mean by "India's semiconductor market"

Three different numbers are often quoted as if they describe the same thing. They do not.

1. **Semiconductor consumption market** – the value of chips consumed by electronics manufacturers in India, i.e. chips that go into products *made in India*, including those later exported. This is the definition used in this report. IESA puts it at **US$52 billion in FY2024-25**.

[1] SIA press release, 1 Feb 2016 (Ref. 23); WSTS Spring 2026 (Ref. 22).

2. **Electronics production** – the value of finished electronic goods made in India. MeitY reports ₹11.3 lakh crore for FY2024-25, or about **US$119 billion**. Chips are only one input here; displays, batteries, PCBs, mechanical parts and assembly also add value.
3. **Semiconductor imports** – the value of loose chips that cross the border under HS 8541 and 8542. This was **US$30.3 billion in FY2024-25** (TradeStat).

The consumption market is roughly 44% of electronics production value, which is reasonable because chips are the single most expensive part of most devices. Imports are smaller than the market because a large share of chips enter India already mounted on circuit boards and inside modules, which are recorded under other HS codes.

3.2 How big is the market? Comparing the estimates

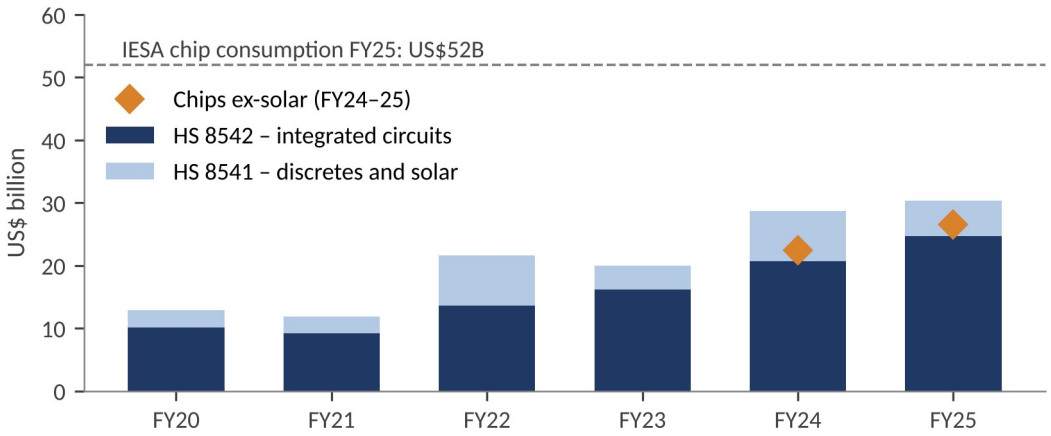
Published estimates range from US$45 billion (IESA–Counterpoint, 2023) to US$64 billion (EY–IESA, 2026), mainly because of base year and definition, and agree on about US$100–120 billion by 2030. EY's 2026 figure is an old projection rather than a measured value, and IESA's 13% growth rate is tied to its older 2023 base (Annex A.2, Exhibit A1).

For the main analysis I use the IESA FY2024-25 consumption estimate of US$52 billion. It has a clear definition, a published bottom-up method, it is the most recent value that is an estimate rather than a forecast, and it sits inside the government's own US$45–50 billion range for the same year.

3.3 Cross-check with import data

TradeStat data shows India's imports under HS 8541 and 8542 rising from **US$12.8 billion in FY20 to US$30.3 billion in FY25**, about 19% a year. Integrated circuits (HS 8542) grew from US$10.1 billion to US$24.7 billion over the same period, about 20% a year, and now make up 3.4% of India's total imports.

Exhibit 2: Loose-chip imports have more than doubled since FY20 but still cover only about half of what India consumes



Source: TradeStat, Ministry of Commerce, HS 8541 and 8542 (Ref. 11); IESA, Feb 2025 (Ref. 28); solar split via busy.in. US$ as reported.

HS 8541 also includes solar cells and modules, which are not part of the electronics chip market; they explain why 8541 jumps around (US$7.9 billion in FY22, US$3.8 billion in FY23, US$8.0 billion in FY24). In FY25, solar was about 67% of HS 8541 imports, or around US$3.8 billion. Removing it, **imports of chips for electronics were about US$26.5 billion in FY25**, up roughly 18% on FY24.

That leaves a gap of about US$25.5 billion against the US$52 billion market. My estimate is that chips inside imported sub-assemblies, valuation differences and classification explain US$15–25 billion, covering the gap only at its top end (Annex A.2, Exhibit A2). Either way, India's real chip import dependence is close to 100% of the market, not the 50–60% suggested by HS 8541/8542 data alone, and with exports of only US$1.6 billion (mostly solar), the chip trade deficit is close to the full import value.

3.4 Where the demand comes from

IESA's bottom-up data shows that mobile handsets are by far the largest user, at about **49% of chip demand in 2023**, rising to **56% by 2030**. By 2030, IT is expected at 12%, industrial at 9% and automotive at 8%. Handsets, IT and telecom together made up more than 75% of demand in 2023. EY–IESA (2026) gives a very different split from a different source, so I do not mix the two (Annex A.2).

Handsets are actually growing faster than the overall market: India was expected to make nearly 300 million phones in 2025, one in four exported. Automotive, industrial and data centres grow fast in percentage terms but from a smaller base.

3.5 Forecast to 2030

Exhibit 3: In the base case India's chip demand nearly doubles to about US$96 billion by 2030

Scenario	CAGR	Reasoning	2030 market
Conservative	9%	India grows only in line with the normal global chip market (7–9% a year, Section 2)	~US$80B
Base	13%	IESA's own growth rate, supported by continued electronics PLI momentum	~US$96B
Optimistic	18%	Close to the 2021–26 growth implied by IESA data and to Deloitte's ~20%	~US$119B

Source: IESA FY2024-25 base (Ref. 28); own scenarios. Workbook, Market sizing tab.

The calculation is simple: 2030 value = 52 × (1 + CAGR)^5. For example, 52 × 1.13^5 = 52 × 1.842 = US$95.8 billion. My base case is below IESA's US$103.4 billion because FY25 came in slightly below IESA's own 13% path from 2023. As a check, loose-chip imports grew about 19% a year over FY20–FY25, above all three scenarios; part of that is phones moving from importing finished devices to importing chips for local assembly, so it overstates how fast chip demand itself grew. The 2026 memory price boom (Section 2) will push the market's dollar value up in 2026 without more chips being used. I treat this as a one-off.

3.6 How much demand is met locally?

Today, almost none of the demand is met by chips fabricated in India. There is no commercial fab yet; SCL Mohali is a small, older-node government facility. In packaging, three units started commercial production in 2026: Micron (February), Kaynes (March) and CG Semi (July). So in FY25, local supply was effectively zero.

Estimating 2030 output only from the 12 approved projects:

- **Fab (Tata–PSMC, Dholera):** 50,000 wafers per month. At the peer median of US$2,208 per wafer and 95% yield, that is about US$1.26 billion a year at full output (Section 8). By 2030: US$0.19–0.50 billion, with a middle estimate of US$0.32 billion.
- **Merchant OSAT/ATMP units (Tata Assam, CG, Kaynes, HCL–Foxconn and smaller units):** about ₹47,183 crore of capex. Using CG Semi's revenue target as a guide (revenue of roughly 0.625× capex at full ramp), full-ramp revenue would be about US$3.1 billion. At 40–80% of ramp by 2030, that is US$1.24–2.48 billion.
- **Micron Sanand:** captive, so not valued here (see Sections 7 and 9.1).

On a revenue basis, Indian facilities would be about US$1.4–3.0 billion in 2030, or 1.5–3.1% of the base-case market (base US$2.2 billion, 2.3%). Locally fabricated chips stay under 1%. These are revenues, not value added: value added is lower because about 55% of an OSAT's cost is imported materials, while the value of the chips is higher because an OSAT earns only a packaging fee. Section 9.1 adds Micron back and gives three measures of local supply.

What this means. India is a large and fast-growing chip *consumer*, about 6–8% of global demand (US$52 billion against US$795.6 billion in 2025 or US$630.5 billion in 2024). But by 2030 almost all the chip value it uses will still be imported. The realistic near-term gain is packaging capacity and some mature-node output, not self-sufficiency. This supports the starting hypothesis.

4. Government policy and incentives

Key takeaways

- Semicon 1.0 paid about 70% of project cost on paper (about 65% in present value). It worked: 12 projects worth ₹1.64 lakh crore are approved and three plants are producing.
- Semicon 2.0 (July 2026) adds materials, equipment and design support, but cuts manufacturing support to 25–40% of eligible capex.
- Money flows slowly. Only about 7% of the Semicon 1.0 outlay was spent by 2025-26.

4.1 Why the policy exists

Section 3 showed that India uses about US$52 billion of chips a year and imports almost all of it. Even with every approved project running, Indian fabs and OSATs would earn only about 1.5–3.1% of the market by 2030. No company builds a fab or OSAT in a new location on market terms when Taiwan, Malaysia and Vietnam already have suppliers, trained staff and lower risk. The government's answer has been to pay a large share of the capital cost.

4.2 Semicon 1.0 and Semicon 2.0

Semicon 1.0 was launched in December 2021 with an outlay of ₹76,000 crore. Support is paid on a *pari-passu* basis: the government releases its share only as the company puts in its own. States add their own support on top; Gujarat gives **40% of the central assistance**, about 20% of project cost. The Design Linked Incentive (DLI) scheme added up to 50% of design costs (capped at ₹15 crore) and 6–4% of sales for five years. The Cabinet approved Semicon 2.0 on 15 July 2026 with **₹1,27,500 crore**, notified on 31 August 2026, with six pillars: design, machines and materials, more fabs, ATMP/OSAT, R&D, and talent.[2] Exhibit 4 sets the two side by side; other sections refer back to it.

Exhibit 4: Semicon 2.0 is a bigger scheme, but it pays a lower share of a smaller cost base for manufacturing

Item	Semicon 1.0 (2021–26)	Semicon 2.0 (from July 2026)
Outlay	₹76,000 crore	₹1,27,500 crore (not split by pillar)
Silicon fab	50% of project cost	40% of eligible capex
Compound / display fab, advanced packaging	50% of capex	35% of eligible capex
Legacy packaging (ATMP/OSAT)	50% of capex	25% of eligible capex
Cost base	Project cost / capex	Eligible capex: excludes land, interest during construction (IDC) and technology transfer
With Gujarat's 40% top-up	~70% on paper, ~65% in present value	Fab up to ~56% of eligible capex (~45% of total); legacy OSAT ~35%
New areas	DLI for design	Materials and equipment (30% capital support), design equity and royalty funding, R&D and talent (detail in Annex B.2)

Source: ISM scheme guidelines (Refs. 2–4); PIB and gazette (Ref. 1); Gujarat Semiconductor Policy 2022–27 (Ref. 12). Present value: Gujarat's 20% is paid in five yearly instalments, worth about 15% today at a 10% discount rate.

For both flagship projects, Micron and Tata's Dholera fab, government pays about 70% and the company about 30% on paper, closer to 65% in present value (worked examples in Annex B.2). The actual payout also depends on the cost finally approved as "eligible".

By July 2026, 12 manufacturing projects with about **₹1.64 lakh crore** of investment were approved: one silicon fab, one SiC fab, one GaN display fab and nine packaging units. On the design side, 24 projects were approved and 105 start-ups were given access to EDA tools. Committed and spent money are very different. The centre has committed roughly ₹82,000 crore (50% of ₹1.64 lakh crore), more than the original outlay, but actual

[2] PIB PRID 2284784, 15 Jul 2026; gazette notification, 31 Aug 2026 (Ref. 1).

spending was ₹681 crore in 2023-24, ₹638 crore in 2024-25 (23% and 9% of budget) and an estimated ₹4,300 crore in 2025-26: about ₹5,600 crore, or 7% of the outlay, because money is released only as plants are built.[3]

The scheme documents set no node or market target; the targets often quoted come from speeches and roadmaps (Annex B.2).

4.3 Does the subsidy change the economics?

For a fab, depreciation is the biggest cost. As an illustration, take the ₹91,526 crore Dholera fab making 50,000 wafers a month, depreciated over 10 years, and compare depreciation with the peer median wafer price of US$2,208 (Section 8).

Exhibit 5: Only Semicon 1.0-level support brings a new fab's depreciation down to an established foundry's level

Capex support	Depreciation per wafer	Share of the US$2,208 peer median price
None	~US$1,600	~72%
50%	~US$800	~36%
Up to 56% (Semicon 2.0 + Gujarat)	~US$700	~32%
70% (Semicon 1.0 + Gujarat)	~US$480	~22%

Source: own calculation; 10-year blended life, 600,000 wafers a year. UMC benchmark: depreciation 21% of revenue in 2025 (Ref. 18).

Foundries usually depreciate equipment over about 5 years, so early-year figures would be close to double, but the pattern holds: at 56% support depreciation is still about 32% of the price, and only at about 70% does it fall to UMC's level. For an OSAT the same logic applies on a smaller scale: at 70% support, depreciation falls to about 8% of revenue, close to Amkor's 8.7%.

4.4 How generous is India, and is it enough?

Semicon 1.0's ~65–70% support is above Japan's for TSMC's first Kumamoto fab (35–40%) and above the US; Semicon 2.0's lower rates bring India closer to other countries (Section 5.2). Gujarat also subsidises power, water, desalination and land, and about 67,000–70,000 students have been trained on EDA tools against a target of 85,000 (Annex B.2).

The policy has achieved its first goal: real factories now exist. But four risks remain. **Delivery is slow:** only about 7% of the outlay had been spent by 2025-26, the first fab is now expected in 2028, and project tenures are being extended. **Projects depend on the subsidy:** lower Semicon 2.0 rates will test whether new projects work at about 35–56% support on eligible capex. **Upstream support started late:** materials and equipment are only now supported, and suppliers take years to build. **Centre and states compete:** states top up at different rates (Gujarat and Assam 40%, Odisha 50%), and Gujarat hosts most projects.

What this means. India now has a packaging base and one fab under construction. Building a full ecosystem will need steady funding for at least another decade and genuine local demand from Indian design firms.

5. How India compares with other chip hubs

Key takeaways

- China, Taiwan and Korea hold about two-thirds of global packaging capacity. India does not appear in any global dataset yet.
- Semicon 1.0 is the most generous support in the comparison. Semicon 2.0 brings India down to roughly the top of the US range.
- Labour is worth only about ±2–3% of an OSAT's sales; the subsidy is worth about 18 points of revenue. By 2030 India will have about 1–1.6% of global packaging, roughly a fifth of Malaysia's share today.

[3] PRS Demand for Grants analysis, MeitY 2026-27 (Ref. 10).

5.1 Where capacity sits today

The latest region-by-region split of manufacturing capacity is still the SIA–BCG 2024 report, with 2022 data; I did not find a newer one. Three countries dominate. China, Taiwan and Korea hold about two-thirds of global packaging capacity (30%, 27% and 9%) and accounted for 79% of equipment spending in 2025.[4] Taiwanese firms alone earned 54% of the revenue of the top ten OSATs in 2024, mostly through ASE; none of the top ten is from Malaysia, Vietnam or India.[5] Malaysia holds about 7% of packaging capacity and Vietnam under 1%. India does not appear separately in any global capacity dataset, so for India I build proxies (Section 5.4). The full country table is in the workbook (Competitive benchmarking tab).

5.2 How generous is India's support?

Taiwan and Korea mainly use tax credits, China state equity funds, Malaysia capital grants, and India a direct cash grant during construction (Annex B.4). The US is the clearest comparison because each award is public. A GAO review of 40 CHIPS projects found grants worth a median of 14% of projected capex; for the biggest awards the grant share runs from 9% (Intel) to 20% (Amkor) (Annex B.4, Exhibit A6).[6] Adding the 25–35% US investment tax credit takes total federal support to at most about 35–55% of capex.

What this means. Semicon 1.0, at about 65–70% of project cost, is more generous than the US, Japan or anyone else in this comparison. Semicon 2.0 (up to about 56% of eligible capex for a fab, with Gujarat's top-up) brings India down to roughly the top of the US range. India was paying a premium to get started. The new scheme assumes it now needs less of one.

5.3 Malaysia and Vietnam: India's real competitors

India is not competing with Taiwan for leading-edge fabs. For packaging, the real competitors are Malaysia and Vietnam. **Malaysia** has done OSAT since 1972, holds about 7% of global ATP capacity, has local test and automation suppliers, and its 2024 strategy (at least RM25 billion of support) admits it is "over-concentrated" in OSAT and wants to move into design and advanced packaging. **Vietnam** is at a similar stage to India but further ahead: it hosts Intel's largest assembly-test plant and a US$1.6 billion Amkor plant, and SIA–BCG expects its ATP share to rise from under 1% to about 8% by 2032 (country detail in Annex B.3).

Exhibit 6: India beats Malaysia and Vietnam on subsidy and home market, and loses on suppliers and customers

Factor	India vs Malaysia	India vs Vietnam
Labour (median factory worker, JETRO FY2023)	Cheaper (US$294 vs 403)	Dearer (US$294 vs 251)
Capex support	Much higher	Much higher
Supplier base, logistics	Weaker	Similar or weaker
Existing customers	Few	Fewer
Domestic chip demand	Much larger (US$52B)	Larger

Source: JETRO Survey on Japanese-Affiliated Companies FY2023 (Ref. 30); Sections 3–4; MITI National Semiconductor Strategy (Ref. 14); Vietnam Decision 1018 (secondary summaries).

What this means. Labour is not the deciding factor. Labour is about 10% of an OSAT's sales (Amkor), so India's labour gap is worth only about 2–3% of sales either way. The capex subsidy matters about six times more: it can move depreciation by about 18 percentage points of revenue (Section 4.3). India's real advantages are the subsidy and its home market. Its real disadvantages are a missing supplier base and no track record with global customers.

5.4 India's realistic share of global packaging by 2030

India has no published share, so I estimate one from the approved projects in two simple ways: by units (about 25.2 billion units a year at full ramp against about 1.44 trillion globally in 2030) and by value (about US$4.6 billion of full-ramp revenue against a global ATP market of about US$156 billion). Method in Annex A.6.

[4] SEMI equipment billings by region, 7 Apr 2026 (Ref. 27).

[5] TrendForce top-10 OSAT ranking, 13 May 2025 (Ref. 31).

[6] GAO-26-107882, Jul 2025 (Ref. 13); US Department of Commerce award releases.

Exhibit 7: Both proxies put India at about 1–1.6% of global packaging in 2030

Scenario	Assumption for 2030	Share by units	Share by value
Conservative	30% of approved capacity running	0.5%	0.9%
Base	55% running	1.0%	1.6%
Optimistic	80% running + 25% more capacity from Semicon 2.0	1.7%	2.9%

Source: own calculation; approved capacity from PRS and PIB (Annex B.5, Exhibit A7); global units from TechInsights; global ATP value from SIA–BCG (Ref. 24).

The two methods weight Tata Assam and Micron differently but land in the same place.

What this means. In the base case, India will have about 1–1.6% of global packaging by 2030. That is roughly a fifth of Malaysia's share today, and it puts India roughly where Vietnam was before its current investment wave. On Criterion B, India will be a credible new entrant in packaging by 2030, not yet a major hub. On fabs, Tata's 50,000 wafers a month is about 0.45% of global 300 mm capacity at full build. It will be well below that in 2030.

6. The wider Indian ecosystem

Key takeaways

- Twelve approved units worth ₹1,64,293 crore, but Tata and Micron are 86% of the money and Gujarat hosts 78%.
- Every merchant packaging unit that is running has a foreign chip company as its first customer.
- India has about 20% of the world's chip designers but Indian design firms earn under ₹150 crore. Equipment and materials, 17% of industry value, are almost absent.

6.1 The 12 approved projects

Semicon 1.0 approved 12 units worth ₹1,64,293 crore (PRS figures), which matches the government's figure of over ₹1.64 lakh crore. The full tracker is in Annex B.5 (Exhibit A7) and the workbook. Three patterns stand out:

- **Concentration:** Tata's two projects are 72% of the money, and Tata plus Micron make up 86%. Units located in Gujarat account for 78%.
- **Mostly packaging:** nine of the 12 are packaging or ATMP units. There is one silicon fab (Tata–PSMC, Dholera) and two compound-semiconductor fabs (SiCSem for SiC power chips; Crystal Matrix for GaN micro-LED displays).
- **Early stage:** three units are in commercial production: Micron (February 2026), Kaynes (March 2026) and CG Semi (July 2026). Tata Assam began pilot production in October 2026. The Dholera fab and HCL–Foxconn's display-driver unit are due in 2028. The status of the four units approved in August 2025 is not confirmed.

Who makes what, for whom. Every merchant unit that is running has a foreign chip company as its first customer (CG Semi: Renesas; Kaynes: Alpha & Omega; Tata Assam: Qualcomm and Nexperia; unit-by-unit detail in Annex B.5). This is how OSATs normally start, but it also means India's packaging volumes depend on decisions made abroad.

6.2 Design

India has about 20% of the world's chip-design engineers but captures little of the design revenue: Indian design companies earn under ₹150 crore in total. The reason is ownership. Most engineers work in captive centres of Intel, Qualcomm, AMD, NXP and others. These centres are paid on a cost-plus basis, while the chip IP and product revenue stay with the parent abroad. Indian product start-ups face long development cycles, high EDA and tape-out costs and little venture funding. DLI approved 24 of 115 applications (21%) and gave EDA access to 105 firms, against a target of 100 funded companies; Semicon 2.0's equity and royalty funding suggests the government also saw grants alone as insufficient.

6.3 Why every stage matters, and where the gaps are

By value added, design is 56% of the industry, fabs 19%, equipment 12%, materials 5% and packaging only 6% (Exhibit 1). India is building its presence in the smallest slice: there is no commercial fab until Dholera ramps (at 28–110 nm), packaging is still ramping and mostly legacy, there is no large domestic supply of wafers, photoresists or specialty gases, and equipment is almost entirely imported. Packaging alone also brings limited local value: materials are about 55% of an OSAT's cost (Amkor), and almost all of them are imported today. So each rupee of OSAT revenue in India adds much less than a rupee of domestic value.

What this means. India's biggest gaps are equipment and materials, together 17% of value and almost nothing in India; they are only now supported under Semicon 2.0. The next gap is owned design IP, where the talent exists but the companies do not. Packaging is the right place to start, but the share of value India captures will rise only if suppliers and Indian product companies grow alongside it.

7. Case study: Micron's Sanand assembly and test plant

> **Key takeaways**
>
> - Micron came to India mainly because of China risk and a 70% subsidy. The Indian market was the weakest reason.
> - Sanand opened on schedule in February 2026. In the base case about US$4.7 billion of memory passes through it in 2030.
> - Most of that is exported and the wafer is imported, so Sanand adds only about 1% of India's market in local supply and 0.26% in value added.

Micron is the world's third-largest memory maker, with 22–23% of DRAM revenue in 2026.[7] Sanand is its first plant in India and India's first commercial chip plant of any kind. It does not make chips. It takes finished DRAM and NAND wafers from Micron's fabs in Taiwan, Singapore, Japan and the US and turns them into memory chips, modules and SSDs. It is captive (Section 1), so this section looks at what Sanand does and what it adds, not at its profit.

7.1 Why India, and why in 2023?

Ranked by how strong the evidence is (detail and dates in Annex C.1):

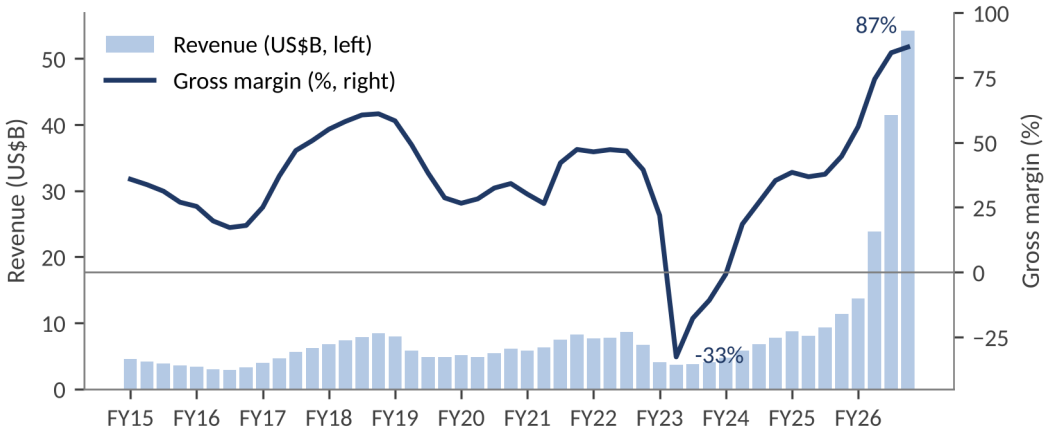
1. **China risk (strongest).** China's May 2023 ban on Micron in critical infrastructure was followed, six days after Micron flagged a "low-double-digit" revenue risk, by the India announcement on 22 June. China-headquartered customers fell from about a quarter of revenue to 11% in FY2026, though Micron also added about US$600 million at Xi'an: diversification, not an exit.
2. **Subsidy (strong).** The centre and Gujarat pay 70% of the project; Micron's share is up to US$825 million (Section 4.2).
3. **Back-end capacity (medium).** New back-end space in Taiwan and Singapore is going to HBM for AI, so conventional assembly needed a new home.
4. **Cost and talent (medium).** A factory worker costs about US$294 a month in India, against US$403 in Malaysia and US$510 in China.
5. **India's own market (weakest).** The first shipment went to Dell in India, but the site serves the world and its capacity already exceeds India's memory needs.

What the plant does. The wafer is thinned and cut into dies. The dies are stacked and sealed into packages, tested, and then mounted onto memory modules or built into SSDs. Micron keeps this in-house because memory testing is part of its product edge: it designs its own test hardware, and test results feed back into fab yields. Sanand is built for conventional packages (BGA chips, modules, SSDs), not HBM.

[7] TrendForce DRAM market-share reports, 4Q25–2Q26 (Ref. 31).

7.2 The memory boom and what it means for Sanand

Exhibit 8: Micron's gross margin swung from −33% to +87% in three years, so Sanand's volumes will follow a very cyclical parent



Source: Micron 10-K and 8-K filings FY2015–26 (Ref. 15); FY2025–26 quarterly rows via Bigdata.com (FMP data), checked against 10-K totals.

Micron's gross margin fell from about 47% in 2021–22 to −33% in early 2023, then rose to 87% in the last quarter of FY2026, on FY2026 revenue of US$133.2 billion. The cause is AI demand for HBM and server memory; DRAM contract prices are heading for about six times their late-2025 level by end-2026 (Annex C.1).

For Sanand, the boom has two effects. Micron wants every chip it can ship, so the plant ramps quickly: 12-hour shifts were allowed at the site in August 2026. But Sanand only packages wafers. Its output is capped by how many wafers Micron's fabs can make, and those are growing slowly. And when prices turn, as they did in 2023, the plant will be asked to slow down too.

7.3 Ramp-up so far and to 2030

Sanand went from approval (June 2023) to ground-breaking (September 2023), pilot tool installation (February 2024), pilot production (December 2025) and commercial production with first modules to Dell (February 2026), in under three years (milestone table in Annex C, Exhibit A8). Micron guides to "tens of millions" of chips in 2026 and "hundreds of millions" in 2027.

Phase 1 has more than 500,000 sq ft of cleanroom. PIB puts its capacity at about 14 million units a week (~728 million a year). Phase 2 is "similar in scale" and starts in the second half of the decade. The government's project data (PRS) gives 1,352 million units a year. The two figures are not reconciled in any source I found; my reading is that 728 million is probably Phase 1 only and 1,352 million covers both phases. This is still an open item. Against 1,352 million units a year, my 2030 scenarios assume 35%, 55% and 80% utilisation.

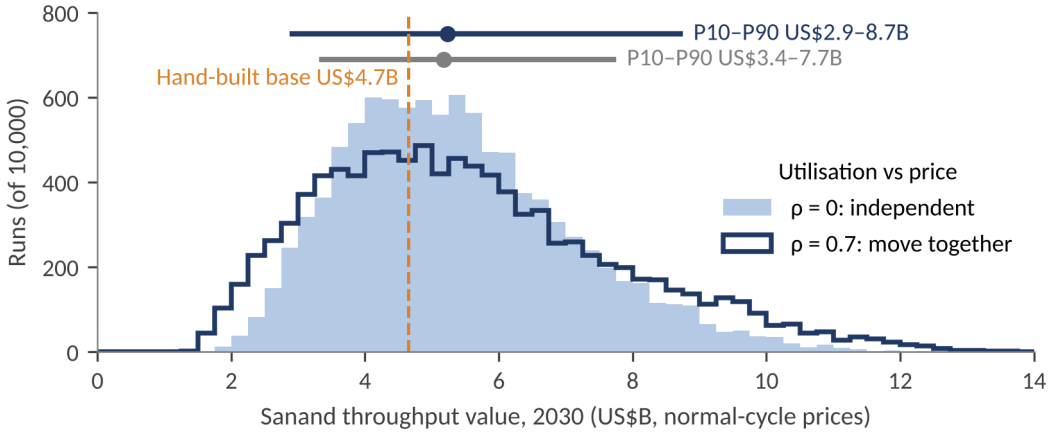
Exhibit 9: In the base case about US$4.7 billion of memory passes through Sanand in 2030, roughly Phase 1 running full

2030	Conservative	Base	Optimistic
Utilisation of 1,352M units	35%	55%	80%
Units	473 M	744 M	1,082 M
Value per unit (normal-cycle prices)	US$4.07	US$6.26	US$8.82
Throughput value	US$1.9 B	US$4.7 B	US$9.5 B

Source: own scenarios; capacity from PRS; prices anchored to TrendForce contract prices, late 2024 to mid 2025 (Ref. 31). Throughput value, not Sanand revenue.

The value uses pre-boom prices (about US$4 per DRAM chip, US$9 per NAND package and US$60 per SSD). A 10,000-run Monte Carlo gives a middle estimate of about US$5.2 billion. If utilisation moves independently of price, the P10–P90 range is US$3.4–7.7 billion; with a 0.7 correlation, as in a real memory cycle, it widens to US$2.9–8.7 billion, which I treat as the realistic band. My first run, with prices drawn independently, gave a narrower US$3.6–7.3 billion (Annex A, Exhibit A4). Utilisation and the NAND price matter most.

Exhibit 10: Correlation leaves Sanand's middle estimate at about US$5.2 billion but widens the range to US$2.9–8.7 billion



Source: own Monte Carlo, 10,000 runs, triangular inputs, one shared memory-price driver (phase7_models.ipynb; chart_sanand_mc_corr.py).

What this means. The base case is roughly Phase 1 running full by 2030. Anything above that needs Phase 2 on time.

7.4 Sanand's place in Micron's network, and what it adds to India

Micron's 10-K lists seven countries with principal plants: front-end fabs in the US and Japan, fabs plus back-end in Taiwan and Singapore, and back-end only in Malaysia, China and India. The 10-K mentions "Gujarat" but never names Sanand, and in four years of earnings calls (FY2023–26) India comes up in 7 passages and Sanand by name in none. India held US$609 million of long-lived assets in September 2026, 0.95% of Micron's total, but those assets are shown after deducting the subsidy.

A press report that Sanand at full capacity would be about 10% of Micron's "global output" is plausible in units (10–14% of a normal year's chips) but not in value (3.5% of FY2026 revenue); Micron itself has never said it (check in Annex C.1).

For India, memory worth about US$4.7 billion passing through Sanand in 2030 equals about 5% of India's US$96 billion chip market. But two things shrink India's real gain. **Most of it is exported:** if 20% is sold in India (my assumption), local supply is about US$0.9 billion, or 1% of the market. **The valuable part is imported:** back-end work is perhaps 10–15% of a memory product's value, and even that includes imported substrates, lead frames and controllers. If about 45% of back-end value stays in India, value added is about US$0.25 billion in 2030 (0.26% of the market). So Sanand raises the share of chips *physically packaged* in India far more than it lowers India's *import dependence*.

On costs, the subsidy mattered to Micron as a saving on the capital it had to put in (US$825 million of US$2.75 billion), not because it changes the cost of a memory chip: depreciation is only about US$0.09–0.29 per unit against a unit value of about US$6.3 (Annex C.1). What India does not get from Sanand is a fab, memory IP or control over where the output goes. The main risks are listed in Annex C.1, Exhibit A9.

What this means. Sanand shows that India can attract a world-class back-end plant and run it on schedule. It is an export platform that adds packaging skills and jobs, not a source of chips that reduces India's import dependence.

8. Case study: Tata Electronics – Dholera fab and Assam OSAT

Key takeaways

- Tata's chip units are pre-revenue and lost ₹776 crore in FY26. The group can fund a long build-out, but its own companies cannot fill the fab.
- Dholera's official commissioning date is 2028. In the middle case it runs at about a quarter of design capacity in 2030 and makes gross-margin losses.
- The binding constraint is customer access and qualification, not the size of Indian demand. Tata also depends on one mid-sized licensor, PSMC.

Tata Electronics is building India's first commercial wafer fab at Dholera, Gujarat, and a large assembly and test plant at Jagiroad, Assam. Together they are worth ₹1.18 lakh crore, about 72% of all approved investment (Section 6). Tata Electronics is unlisted (Section 1), so this section uses project disclosures and peer foundries.

8.1 Why a conglomerate, and can Tata be its own customer?

Tata Electronics had revenue of ₹1,31,082 crore in FY26, mostly from electronics assembly, mainly iPhones.[8] The group's case for chips is strategic: it can fund a long build-out that a stand-alone start-up could not. The semiconductor units are still pre-revenue. In FY26 the Dholera fab company had no turnover, a loss of ₹369 crore and total assets of ₹7,006 crore, under 8% of the ₹91,526 crore project. The Assam unit lost ₹407 crore. Together they explain about half of Tata Electronics' ₹1,611 crore consolidated loss (the loss figure is from RoC filings via ET).

Can the group be a captive customer? Only a little. Apple chooses the chips in Tata's iPhone lines, Tata Motors buys through Tier-1 suppliers, and Tejas Networks (₹1,137 crore revenue in FY26) is small next to a fab worth about US$1.26 billion a year at full load. Group demand can help with early qualification, but it cannot fill the fab.

8.2 Dholera, PSMC and Assam

Dholera is a 300 mm fab for 50,000 wafer starts a month at 28, 40, 55, 90 and 110 nm, licensed from PSMC, for power management ICs, display drivers, microcontrollers and logic, with Nexperia MOSFETs planned. It costs ₹91,526 crore (~US$9.6 billion) with 50% central support, and its official first commissioning date is **2028** (fact table and timeline in Annex C, Exhibit A10). A press report of "trial production" in late 2026 refers to test wafers, not saleable output, so I use PIB's 2028 date (Annex C.2).

Why 28–110 nm? These nodes do not need EUV lithography. PSMC can license them, and they make most power, analog, display-driver and microcontroller chips. These are the chips India's car, industrial and power-electronics sectors use, and the ones in short supply in 2021.

PSMC provides fab design and construction support, a process licence and engineering support for transfer; its decks do not show Tata income. It was loss-making when the deal was signed and recovered on AI-driven tightness in 2026 (Annex C.2). The risk is not that PSMC fails, but that a mid-sized partner (10th among foundries, focused on its own "3D AI foundry" plans) may have its attention elsewhere just when Dholera needs the most help with yield.

Assam. The Jagiroad plant (₹27,000 crore, ~US$2.8 billion, 48 million chips a day) started pilot chips in October 2026, roughly 16–19 months after Tata's original mid-2025 target (a weak source), with Qualcomm, ROHM and Nexperia as named customers (Annex C.2). Assam will reach volume two years before Dholera. It is Tata's first real test of execution and customer qualification.

8.3 Who will buy Dholera's output?

India's demand is not too small. Using the Section 3 base case (US$96 billion in 2030) and assuming 30% is mature-node logic, analog and power, about **US$29 billion** of India's 2030 demand is the kind of chip Dholera

[8] Tata Sons Annual Report 2025-26, including Form AOC-1 (Ref. 16).

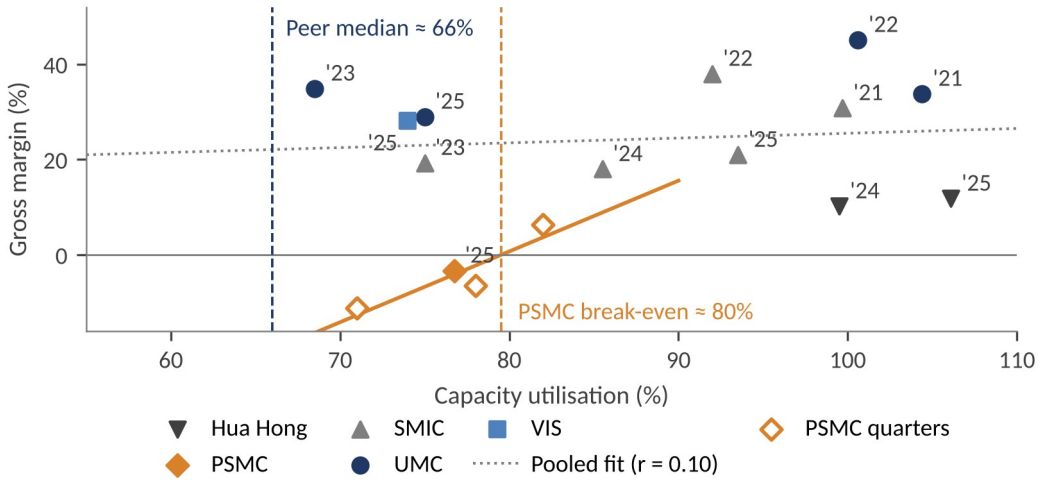
could make. Even at full capacity Dholera would produce about **4%** of that, and in the middle ramp case about **1%**.

The real constraint is who decides. Most chips used in India are chosen by global chip companies (Qualcomm, MediaTek, TI, Infineon, NXP), and automotive qualification takes two to three years, so partners such as Nexperia, Bosch, ROHM, Himax and ADI matter more than the size of Indian demand. If India-designed chips stay few (24 DLI approvals, Section 6.2), Dholera will mostly export into a market where Chinese foundries are adding about 70% of new 12-inch mature-node capacity.[9]

8.4 Economics: what a new mature-node fab looks like

I built a dataset of six mature-node foundries (UMC, GlobalFoundries, PSMC, VIS, SMIC, Hua Hong) from their filings for 2021–25. **Price per wafer:** peers earn US$990–3,340 per 12-inch-equivalent wafer, with a median of **US$2,208**; UMC, the closest match, earned about US$1,970 in 2025. My first market estimate had assumed US$3,000, which was too high; the report now uses the peer median throughout. **Break-even loading:** fixed costs are about 64% of a foundry's manufacturing cost (UMC 20-F), but across companies utilisation alone barely explains gross margin (r = 0.1), because price and fab age differ (Exhibit 11). Estimated company by company, old depreciated fabs (UMC, VIS) break even at about 46%, fabs carrying new depreciation (PSMC, Hua Hong) at 80–88%, and the peer median is about 66%. PSMC's own quarters confirm this: it lost money at 71–78% utilisation and turned positive at 82%.

Exhibit 11: Across peers, utilisation alone barely explains gross margin; fab age and price matter as much



Source: company filings and investor reports 2021–25 (UMC, GlobalFoundries, PSMC, VIS, SMIC, Hua Hong; Refs. 17, 18, 21), Bigdata.com tearsheets; foundry_peers.csv.

Ramp. A 10,000-run simulation of utilisation, yield and price (over the full peer range, boom and slump) gives about 149,000 wafers in 2030 and an illustrative output value of US$0.32 billion (US$0.19–0.50 billion), rising to 380,000 wafers and US$0.81 billion in 2032, against about US$1.26 billion at full capacity (Annex C.2, Exhibit A11).

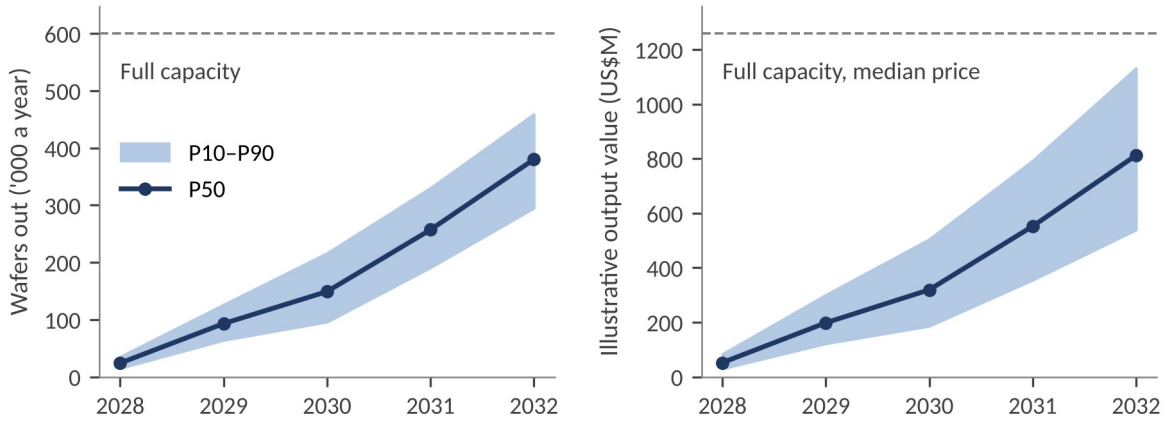
What this means.

- **2030:** in the middle case Dholera produces about a quarter of its design capacity. Peer break-evens (46–88%) are measured on installed capacity, and Dholera adds tools in phases, so this is not like-for-like. But the building, cleanroom and utilities are built for the full fab and depreciate from day one, so on any reasonable view Dholera makes gross-margin losses in 2030. By 2032, assuming full tool-in, 45% of runs are still below the peer median and 87% below the new-fab level. Utilisation, meaning finding customers, moves the result far more than price or yield.
- **The subsidy:** with ~70% support, depreciation falls from about 72% to about 22% of the median price, close to UMC's 21% (Exhibit 5). The subsidy is what turns Dholera from a structurally loss-making fab into one that can break even once it is loaded.

[9] TrendForce, May 2026 (Ref. 31).

- **Capex per unit of capacity:** Dholera costs about US$193k per wafer-per-month, about 16% above UMC's comparable Singapore fab (US$167k); part is scope, part may be a first-fab premium (Annex C.2).

Exhibit 12: Even by 2032 the middle case reaches only about two-thirds of full capacity



Source: own Monte Carlo, inputs drawn independently (phase6_tata.ipynb). Illustrative output value at peer benchmark prices, not a Tata revenue forecast.

8.5 Supply chain and risks

Construction, utilities and people are local. Wafers, lithography and process tools, photoresists, specialty gases and packaging materials will be imported for the rest of the decade. Tata has signed MoUs with ASML, TEL, Merck, Sumitomo, Sojitz, Synopsys and Enomoto, but most say "explore"; only Fujifilm has committed money to a local plant (₹800 crore for materials in Dholera) (Annex C, Exhibit A12).

The biggest risk is customer acquisition and low utilisation (high likelihood, high impact), followed by mature-node price pressure from China and dependence on PSMC (medium likelihood, high impact); yield learning, execution delay, talent and subsidy timing are medium or lower (risk table in Annex C, Exhibit A13). On China, I tested the price risk both ways (Annex C.2): prices were weak in 2024–25 and are rising 5–15% in 2026, but nearly 70% of new 12-inch mature-node capacity is being built in China, so a 2028–30 down-cycle could hit just as Dholera ramps. That is why my price range starts at the 2024 Hua Hong level (about US$990). At the low price, 2030 output value falls by more than half (US$0.13 billion against US$0.30 billion).

What this means. Tata has the money and patience a fab needs, and the subsidy fixes the cost side. The open question is demand: can Tata win and keep foreign customers at mature-node prices set by Chinese capacity? Dholera supports the hypothesis that mature-node fabrication is *possible* for India by 2030, but at low loading and with losses. Being *competitive* is a 2032+ question.

9. Financial analysis

Key takeaways

- In 2030, plants in India make or package 7.1% of India's chip demand, supply 1.7% to Indian buyers and add 1.3% of its value (base case).
- A reference packaging plant earns a 10-year IRR of −1.1% without support and 15.9% with ~65% support. Price is the input that matters most.
- A reference fab cannot break even without support. With ~65% support it breaks even at 60% loading in 2033, but its 20-year IRR is at best 7.0%, and 1.8% under Semicon 2.0's eligible-capex rules.

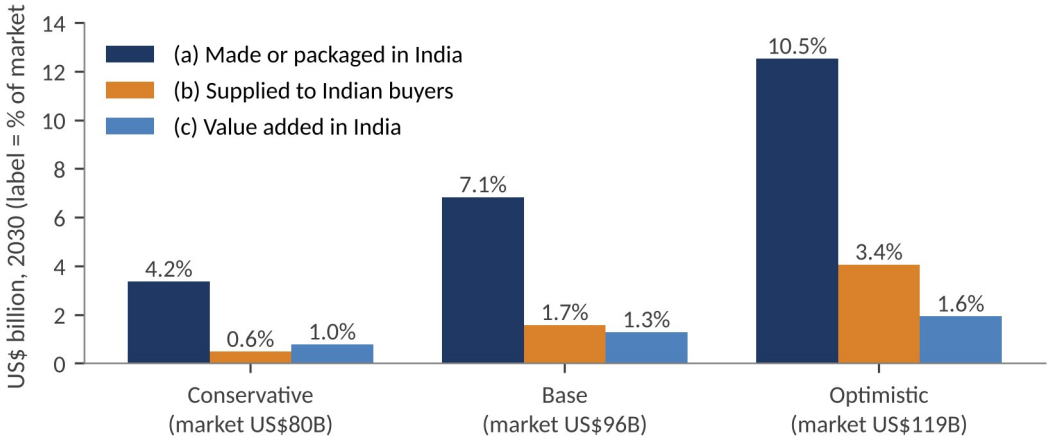
All numbers in this section come from one workbook, *India_Semiconductor_Model.xlsx*, where every assumption sits on the Inputs tab with a source and every result is a live formula. The reference facilities follow the conventions in Section 1; their methods are in Annex A.4. The analysis asks how much of India's 2030 demand will be met locally, whether an Indian OSAT makes money, and whether a mature-node fab does.

9.1 How much of the 2030 market will be local?

I measure "local supply" in three ways, because each answers a different question (formulas in Annex A.3): **(a) made or packaged in India**; **(b) supplied to Indian buyers**, the share of (a) that stays in India; and **(c) value added in India**, revenue minus imported wafers, materials and licences. Merchant OSATs use the tracker's ₹47,183 crore of capex at the 0.625 revenue-to-capex ratio (about US$3.1 billion at full ramp), with 40–80% of full ramp reached by 2030.

Exhibit 13 shows the base case and ranges (exact values in Annex A.3, Exhibit A3). Value added (c) can be larger than value supplied to Indian buyers (b). This happens because value added counts work done in India on exported chips too, while (b) counts only what Indian buyers receive.

Exhibit 13: Most of what is made or packaged in India in 2030 is exported, and most of its value is imported



Source: workbook, Market sizing and forecasts tab; phase7_models.ipynb.

What this means. Even in the optimistic case, India makes or packages only about a tenth of what it consumes, and keeps less than 2% of the market's value. Most of the activity is back-end work on imported wafers. India in 2030 will be a real packaging location, but it will not be self-sufficient. The import bill from Section 3 will keep growing alongside the new plants.

9.2 Reference OSAT: does packaging pay?

The reference plant is the size of CG Semi (₹7,584 crore, 4,044 million units a year) with Amkor's 2025 cost structure, adjusted for Gujarat: labour at about 0.87× and subsidised power at about 0.16× the South-East Asian average (method in Annex A.4). At full ramp it earns about ₹4,740 crore of revenue at a 19.8% EBITDA margin.

Exhibit 14: Without support the reference packaging plant does not pay back in ten years; with ~65% support its IRR is about 16%

Support case (upper bounds)	Break-even utilisation	10-year IRR	Payback
No support	77%	−1.1%	none in 10 years
Semicon 2.0 legacy (35%)	62%	5.4%	year 8
Semicon 2.0 advanced (49%)	56%	9.3%	year 7
Semicon 1.0 central only (50%)	56%	9.7%	year 7
Semicon 1.0 + Gujarat (~65% in PV)	49%	15.9%	year 6

Source: workbook, Financial scenarios tab, block C. Illustrative reference facility, not company economics.

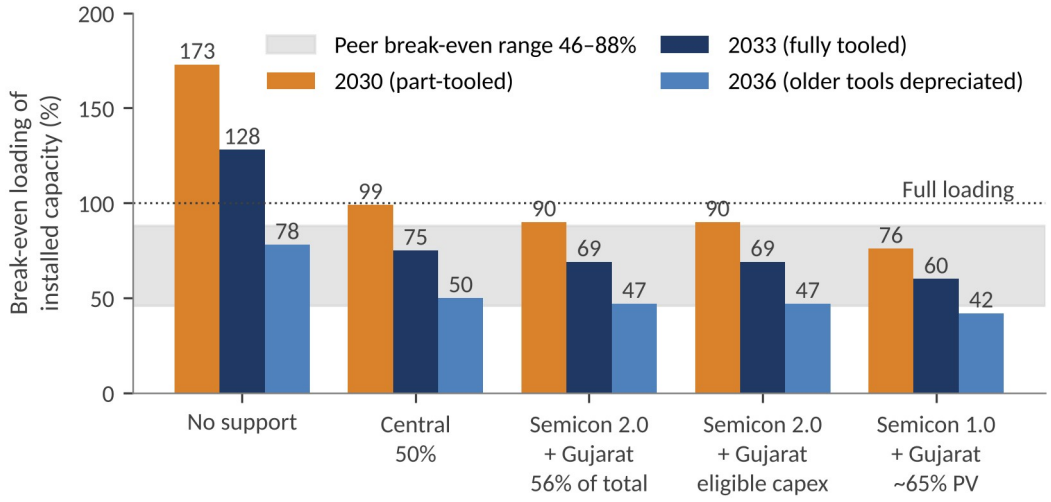
These support cases are **upper bounds**: CG's approved cost may include land and interest during construction, which are not broken out, and Semicon 2.0 does not support those items. The IRR is most sensitive to price. A ±10% change in price per unit moves the IRR from 0.7% to 27.6%, and a −20% price wipes out EBITDA entirely.

What this means. Without subsidy, packaging earns a thin gross margin but is not a viable investment: the 10-year IRR is negative. This is a capital-heavy, low-margin business, even with cheaper labour and power. With Semicon 1.0-level support the plant becomes a reasonable investment. Under Semicon 2.0's lower rates for legacy packaging, the case becomes thin.

9.3 Reference fab: does a mature-node fab pay?

The reference fab is Dholera-sized (₹91,526 crore, 50,000 wafers a month), tooled in phases from 2028 to 2033, priced at the peer median of US$2,208 and costed from UMC's 2025 accounts. Following SIA–BCG, tools are 50% of capex (5-year life), the building 30% (20 years), and land, IDC and licence 20% (method in Annex A.4).

Exhibit 15: Without support the reference fab cannot break even even at full loading; with ~65% support it breaks even at 60% in 2033



Source: workbook, Financial scenarios tab, block D; phase7_models.ipynb. Illustrative reference facility (Dholera-sized), not company economics.

With ~65% support the fab breaks even at 60% loading in 2033, inside the 46–88% peer range (Section 8.4). Under Semicon 2.0's eligible-capex rules, net capex rises by about US$1.1 billion and the 20-year IRR falls from 4.4% to 1.8%; break-even loading does not move, because the excluded 20% (land, IDC, licence) is never depreciated. Across the peer price range (US$992–3,343) the 12-year IRR runs from −26.7% to 13.9%. UMC's R&D and SG&A of 11.3% of revenue is probably low for a new entrant, so the true IRRs are likely lower than shown.

Exhibit 16: Support cuts the fab's break-even loading, but its return stays below a normal hurdle rate in every case

Support case	Break-even 2033	Net capex (US$M)	IRR 2026–37	IRR, 20-year life
No support	128%	9,634	−11.2%	−3.8%
Central 50% only	75%	4,817	−3.6%	3.0%
Semicon 2.0 + Gujarat, 56% of total (upper bound)	69%	4,239	−1.9%	4.4%
Semicon 2.0 + Gujarat on eligible capex (~45% of total)	69%	5,318	−5.2%	1.8%
Semicon 1.0 + Gujarat (~65% in PV)	60%	3,372	1.4%	7.0%

Source: workbook, Financial scenarios tab, block D. Illustrative reference facility, not company economics.

What this means. A mature-node fab in India only works with very high support. Even then, its return over 20 years is in single digits, below a typical 10–12% hurdle rate for a capital-heavy manufacturing project in India (my assumption). This supports my starting hypothesis. Fabrication is the hardest part of the chain for India, and Dholera should be judged as a strategic and ecosystem investment rather than a stand-alone commercial one.

9.4 Uncertainty and limitations

Monte Carlo ranges for Sanand and Dholera are quoted at two correlation settings, ρ = 0 and ρ = 0.7 (Section 7.3 and Annex A, Exhibit A4). Correlation leaves the middle value almost unchanged but widens the range, so planning should use the ρ = 0.7 band. The main limitations (peer-based cost structures, CG-based revenue ratios, assumed domestic and value-added shares, and an IRR that excludes ecosystem value) are set out in Annex E.

10. Strategic recommendations

Key takeaways

- Prioritise packaging now and Indian-owned design IP for the long term. Do not add a second fab before Dholera proves it can be loaded.
- Utilisation and price are the real levers, so support should reward customers and output, not only capex; Semicon 2.0 may be too thin for new fabs (20-year IRR ~1.8%).

Each recommendation runs: evidence → so what → recommendation → one metric to track.

10.1 Which parts of the value chain should India prioritise?

- **Evidence:** packaging is 6% of industry value and design 56% (2.3); India reaches ~1–1.6% of global packaging (5.4); the reference packaging plant returns 15.9% with ~65% support against at best 7.0% for a fab (9.2, 9.3); only 1.3% of 2030 demand is value added locally (9.1).
- **So what:** packaging is the only stage where India can compete by 2030, but on its own it adds little value. Design is where the value sits.
- **Recommendation:** run two tracks: packaging now, for volume, jobs and supplier pull; Indian-owned design IP for the long term. Treat fabs as a selective, strategic bet.
- **Metric:** India's share of global assembly and test value (base case 1.6% by 2030).

10.2 OSAT, mature-node fab, advanced packaging or design?

- **Evidence:** under Semicon 2.0 a legacy packaging plant returns ~5.4%, an advanced one ~9.3% (9.2), and a new fab ~1.8% over 20 years (9.3); Malaysia calls itself "over-concentrated" in legacy OSAT (5.3); DLI approved 24 of 115 applications and Indian design firms earn under ₹150 crore (6.2).
- **So what:** legacy packaging is thin at the new rates; advanced packaging earns more support and value per unit; design has the most value but few Indian product companies.
- **Recommendation:** for new approvals, favour advanced packaging over legacy packaging. Back design start-ups with equity and early customers, as Semicon 2.0 now allows. Hold back on a second fab.
- **Metric:** share of new packaging approvals (by capex) that are advanced packaging.

10.3 Which end-use markets offer the best opportunity?

- **Evidence:** handsets are 56% of 2030 demand, but their chips are chosen by global firms (3.4, 8.3); mature-node logic, analog and power is ~US$29 billion of 2030 demand, of which Dholera could supply ~4% (8.3); Assam's first customers are automotive and power, and HCL–Foxconn targets display drivers (6.1, 8.2); AI servers are tightening the power-chip market (8.5).
- **So what:** phones are the biggest market but the hardest to win; automotive, industrial and power fit India's nodes and already have customers.
- **Recommendation:** point Dholera and the merchant OSATs at automotive, industrial and power chips, including AI-server power. Use display drivers to tie local packaging to India's phone and TV assembly.
- **Metric:** number of automotive-qualified products shipping from Indian plants.

10.4 What advantages can India realistically develop?

- **Evidence:** labour is worth ±2–3% of an OSAT's sales against ~18 points for the subsidy (5.3), and the subsidy is being cut (Exhibit 4); demand grows from US$52 billion to ~US$96 billion (3.5); India has ~20% of the world's chip designers (6.2); subsidised power costs ~0.16× the regional average (9.2).
- **So what:** cheap labour is not a lasting edge, and the subsidy is temporary. The durable advantages are the home market and engineering talent.
- **Recommendation:** sell India on its market and its engineers, not on wages. Link the electronics assembly base to chips packaged in India.
- **Metric:** share of India's chip demand supplied by Indian plants, measure (b) in Section 9.1 (base 1.7% in 2030).

10.5 What are the main risks?

- **Evidence:** price swings move the packaging IRR from 0.7% to 27.6% and the fab IRR from −26.7% to 13.9% (9.2, 9.3); Micron's margin swung from −33% to +87% (7.2); China is adding ~70% of new 12-inch mature-node capacity (8.3); Tata and Micron are 86% of investment and Gujarat 78% (6.1); ~7% of the Semicon 1.0 outlay was spent by 2025-26 (4.2); Dholera relies on one licensor (8.2).
- **So what:** the risks are commercial more than technical. One delayed flagship would set back the whole programme.
- **Recommendation:** track customer wins and utilisation at each plant, not just construction milestones. Spread Semicon 2.0 approvals across more investors.
- **Metric:** share of approved capex outside the two largest investors (today 14%).

10.6 How should Micron strengthen its Indian operations?

- **Evidence:** Sanand is not named in Micron's 10-K or on earnings calls, and India is 0.95% of Micron's long-lived assets; ~80% of output is assumed exported and only ~45% of back-end value stays in India (7.4); growth beyond the base case needs Phase 2 (7.3); the memory boom is already pushing the ramp (7.2).
- **So what:** Sanand is a low-profile export platform; Micron gains most from diversified back-end capacity at a lower landed cost.
- **Recommendation:** start Phase 2 while demand is strong and the original support terms apply. Qualify Indian suppliers for substrates, trays and consumables. Add module and SSD lines for Indian PC and server makers to grow local sales.
- **Metric:** Sanand unit output against capacity, which would also settle the 728 million vs 1,352 million question.

10.7 What capabilities should Tata Electronics prioritise?

- **Evidence:** Tata's chip units are pre-revenue and lost ₹776 crore in FY26 (8.1); customer access, not Indian demand, is the constraint (8.3); new fabs break even only at 80–88% utilisation (8.4); PSMC is 10th among foundries and focused on its own AI products, and Assam is ~16–19 months late (8.2).
- **So what:** money is not Tata's constraint. Selling foundry capacity and running a fab at high yield are.
- **Recommendation:** build a foundry sales and customer-qualification team aimed at automotive and industrial anchor customers on long-term contracts. Build in-house process and yield engineering to reduce dependence on PSMC. Use Assam to prove execution before Dholera ramps.
- **Metric:** share of Dholera's 2030 capacity covered by signed long-term agreements.

10.8 What policy changes would improve India's competitiveness?

- **Evidence:** on Semicon 2.0's eligible-capex rules a second fab returns ~1.8% over 20 years and needs ~US$1.1 billion more of its own money (9.3); at 56% support fab depreciation is still 32% of the wafer price against UMC's 21% (4.3); a legacy packaging plant earns ~5.4% (9.2); ~55% of OSAT cost is imported materials (6.3).
- **So what:** Semicon 2.0 may be too thin to attract new fabs, and capex grants do not reward the input that matters most, utilisation.
- **Recommendation:**
  - **Fabs:** either keep effective support near Semicon 1.0 levels (about 65%) for a limited number of fabs, or accept that no second fab is likely before 2030.
  - **Support design:** link part of the support to output or qualified customers (paid per unit shipped), not only to capex.
  - **Disbursement:** publish milestone-based payment schedules so money flows as plants are built.
  - **Suppliers:** use Semicon 2.0's 30% materials support to build suppliers inside the existing Gujarat cluster, where demand already sits.
- **Metric:** actual central disbursement as a share of committed support.

11. Conclusion: testing the hypothesis

The evidence supports the starting hypothesis (Section 1) on direction, but not all of its wording.

Exhibit 17: The hypothesis holds on direction, but "competitive" needs softening for packaging and a later date for mature-node fabs

Part of the hypothesis	Verdict	Evidence
Competitive player in assembly, test and packaging	**Partly supported.** A credible new entrant in packaging with ~1–2% global share; plant-level costs competitive only because of subsidies; ecosystem not yet competitive	~1.6% of global ATP in 2030 vs Malaysia ~7% (5.4); IRR −1.1% unsupported vs 15.9% supported (9.2); 1.3% value added (9.1)
Selected mature-node manufacturing	**Supported, with a timing caveat.** Possible, but loss-making until the early 2030s	Dholera ~25% of design capacity in 2030 (8.4); fab breaks even at 60% only by 2033 with ~65% support; 20-year IRR at best 7.0% (9.3)
Selected compound-semiconductor manufacturing	**Too early to judge.** Two units approved, status not confirmed	SiCSem and Crystal Matrix approved in 2025–26; no construction start confirmed (6.1)
Leading-edge fabrication remains difficult	**Strongly supported.** Not realistic by 2030	Dholera is 28–110 nm, no EUV (8.2); Taiwan and Korea hold all sub-10 nm capacity (2.3)

Source: Sections 2–9 as cited.

On the three criteria from Section 1: India's **domestic manufacturing contribution** in 2030 is small (7.1% made or packaged, 1.7% supplied to Indian buyers, 1.3% value added). Its **position in the global supply chain** improves from almost nothing to about 1–1.6% of packaging and about 0.45% of 300 mm fab capacity once Dholera is fully built. Its **manufacturing economics** are competitive at plant level only with subsidy: labour helps by about 2–3% of sales, while support moves costs by about 18 points of revenue.

Answer to the central question. India cannot become a globally competitive chip hub by 2030 in the full sense, but it can become a credible packaging location with about 1–2% of world capacity. Packaging, and in the longer run Indian-owned design, are the most attractive parts of the chain. A mature-node fab is possible but will need high support and loyal customers to break even in the early 2030s. Leading-edge fabrication is not a realistic goal for this decade.

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Annex A. Methodology

A.1 Data sources

Official documents (PIB, ISM, MeitY, PRS, TradeStat, state policies) give approvals, outlays and trade. Company filings (Micron 10-K, Tata Sons annual report, UMC, PSMC, Amkor) give financials and plant facts. Industry bodies (WSTS, SIA, SEMI, IESA, JETRO, TrendForce) give market, capacity and cost data. Each number in the workbook's register carries its value, year, source, link where available, and whether it is actual, estimated, forecast or assumed.

A.2 Market sizing: estimates and the import gap

Two points stand out. First, EY's "US$64 billion in 2026" matches a *forecast* made earlier: the IESA–Counterpoint report of 2022 valued the market at US$27.2 billion in 2021 and projected US$64 billion for 2026, so EY's figure should be read as an old projection, not a measured value. Second, IESA's growth rate is tied to its older base. Going from US$45 billion (2023) to US$103.4 billion (2030) is 12.6% a year, close to the stated 13%. Reaching US$103.4 billion from US$52 billion in five years would need 14.7% a year.

Exhibit A1: Published estimates of India's chip market agree on about US$100–120 billion by 2030 but start from different bases

Source (release)	Base year value	2030 value	Implied CAGR	Definition
IESA–Counterpoint (Oct 2024)	US$45B (2023)	>US$100B	13% (2023–30)	Bottom-up chip BoM across 8 device sectors
IESA (Feb 2025)	US$52B (FY25)	US$103.4B	13% stated	"Consumption" market, same BoM method
Government/MeitY (2024–25)	US$38B (2023); US$45–50B (FY25)	US$100–110B	~16%	"Industry estimates", no method given
Deloitte TMT Predictions (Mar 2026)	Not stated in the press release	~US$120B	~20%	Not defined
EY–IESA (Sep 2026)	US$27B (2021); US$64B (2026)	~US$106B (interpolated on the 2026–35 path, ~13.5%)	15.4% (2021–35)	Not defined; cites NITI Aayog

Source: as listed in the first column (Refs. 28, 29, 32); implied CAGRs are my own calculation.

Demand split. EY–IESA (2026) gives consumer electronics 30%, automotive 16%, industrial 15%, data centres 14% and telecom 13%. This comes from a commercial market-research source with a different category structure from IESA's bottom-up split (handsets 56%, IT 12%, industrial 9%, automotive 8% by 2030), so the two are not mixed.

Gap workings. The US$15–25 billion is made up of chips inside imported sub-assemblies (phone PCBAs, camera, display and memory modules: US$10–18 billion), valuation differences (US$3–5 billion) and classification and timing (about US$2 billion). If IESA also counts chips in imported finished devices, the gap closes more comfortably; this is unconfirmed. Exports under HS 8541/8542 were US$1.6 billion in FY25, most of it solar modules.

Exhibit A2: **Chips inside imported sub-assemblies explain most of the gap between imports and consumption**

Part of the gap	Assumption	Estimate (US$B)
Chips inside imported sub-assemblies (phone PCBAs, camera and display modules, memory modules) used by Indian manufacturers	*Assumed* US$30–40B of such imports, 35–45% chip content	10–18
Valuation difference (customs value vs market price)	10–20% on direct imports	3–5
Classification, SEZ timing, inventory	Residual	~2
Explained under our definition		**15–25**
Memo: chips inside imported finished devices (laptops, servers, telecom gear)	US$15–20B of imports, 25–35% chip content	4–7 (not counted)

Source: own estimates; gap = US$52B consumption − US$26.5B loose-chip imports ex-solar = US$25.5B. Workbook, Market sizing tab.

A.3 Three measures of local supply (Section 9.1)

- *(a) Made or packaged in India* = Micron Sanand throughput value + Dholera illustrative wafer value + merchant OSAT revenue.
- *(b) Supplied to Indian buyers* = Sanand × domestic share (10–30%) + (Dholera + merchant OSATs) × domestic share (20–40%).
- *(c) Value added in India* = Sanand × back-end share (12%) × local share (45%) + merchant OSAT revenue × 45% + Dholera × 60%.

Exhibit A3: **Even in the optimistic case India keeps less than 2% of its 2030 chip market's value**

2030 (US$B)	Conservative	Base	Optimistic
India chip market	80.0	95.8	119.0
(a) Made or packaged in India	3.4 (4.2%)	6.8 (7.1%)	12.5 (10.5%)
(b) Supplied to Indian buyers	0.5 (0.6%)	1.6 (1.7%)	4.1 (3.4%)
(c) Value added in India	0.8 (1.0%)	1.3 (1.3%)	1.9 (1.6%)

Source: workbook, Market sizing and forecasts tab, block 4; inputs from Sections 3, 7 and 8.

A.4 Reference facilities (Section 9)

Both are built from listed peers' cost structures, sized like an approved Indian project, and run with live formulas in the workbook.

- **Packaging plant:** CG Semi size (₹7,584 crore, 4,044 million units a year). Revenue = 0.625 × capex at full ramp (CG Semi's US$500 million target ÷ ₹7,600 crore at ₹95). Costs from Amkor 2025, with labour and power adjusted using JETRO FY2024 city data. Ramp 30% / 55% / 75% of capacity in years 1–3, then 85%. Depreciation over 7 years on capex net of support. IRR on 10 years of pre-tax cash flow (no tax, working capital, replacement capex or terminal value).
- **Fab:** Dholera size (₹91,526 crore, 50,000 wafers a month). Capex split from SIA–BCG: tools 50% (5-year life from the year installed; sensitivity 6 years and 50–65% equipment share), building 30% (20 years), land, IDC and licence 20% (not depreciated). Installed capacity 5,000 / 15,000 / 20,000 / 30,000 / 40,000 / 50,000 wafers a month in 2028–33; loading 50% rising to 85%; yield 75% rising to 95%. Costs from UMC 2025: 63.9% of manufacturing cost fixed. Break-even loading = (fixed cash cost + depreciation) ÷ (contribution per wafer × installed wafers). IRR over 2026–37 with end book value, and over a 20-year life with sustaining capex of 10% of revenue. The eligible-capex toggle changes net capex, IRR and payback but not break-even loading.

A.5 Monte Carlo method

10,000 runs with triangular distributions (low, most likely, high) and a fixed seed (42). For correlated runs, two normal numbers are drawn with correlation ρ ($z2 = \rho \cdot z1 + \sqrt{(1-\rho^2)} \cdot e$), turned into uniforms with the normal CDF, and fed into the same triangular distributions (a Gaussian copula). For Sanand the three memory prices share

one price driver. I report ρ = 0 and ρ = 0.7. The first Sanand run drew each memory price independently and is quoted only with that label.

Exhibit A4: Letting utilisation and price move together widens both ranges but barely moves the middle

2030	ρ = 0 (P10 / P50 / P90)	ρ = 0.7 (P10 / P50 / P90)	Prices drawn independently
Sanand throughput value (US$B, captive, not revenue)	3.4 / 5.2 / 7.7	2.9 / 5.2 / 8.7	3.6 / 5.2 / 7.3
Dholera illustrative output value (US$B)	0.19 / 0.32 / 0.51	0.16 / 0.32 / 0.58	0.19 / 0.32 / 0.50

Source: phase7_models.ipynb (montecarlo_correlated.csv); phase5_micron.ipynb and phase6_tata.ipynb for the independent runs. Dholera 2032 at ρ = 0 / 0.7: US$0.54–0.81–1.14B / US$0.48–0.82–1.25B.

A.6 India's share of global packaging (Section 5.4)

India has no published share, so I estimate one from the approved projects in two simple ways. **By units:** the ten approved packaging units (the nine packaging units, including Micron, plus SiCSem's packaging line) add up to about 25.2 billion units a year at full ramp; Tata Assam alone is 62% of this. Global output in 2030 should be about 1.44 trillion units (TechInsights' O-S-D forecast, assuming ICs are about 32% of all units). **By value:** the same ten units cost ₹69,699 crore. Using the ratio from Section 3.6 (revenue ≈ 0.625 × capex), their full-ramp value is about ₹43,600 crore, or US$4.6 billion. Global ATP was about US$95 billion in 2023; growing at 7.3% a year, it reaches about US$156 billion in 2030. The unit count is a rough proxy: a simple TO-package diode counts the same as a memory package, so Tata Assam's high volume of simple packages pushes the unit share up, while Micron's high-value memory pushes the value share up.

Annex B. Industry, policy and ecosystem detail

B.1 Segment economics

Exhibit A5: Foundries and packagers spend heavily on capex; fabless firms spend on R&D instead

Model	Gross margin	R&D, % of revenue	Capex, % of revenue	Example, 2025
IDM	52%	14%	20%	Micron FY26: GM 81% (FY23: −9%)
Fabless	50%	20%	4%	Qualcomm FY25: GM 55%, capex 3%
Foundry	40%	9%	34%	TSMC: GM 60%, capex 33%; UMC: GM ~29%
OSAT	17%	4%	16%	Amkor: GM 14%, capex 13.5%

Source: SIA–BCG (2021), averages for 2016–19 (Ref. 25); company filings for 2025 examples (Refs. 15, 18, 19).

B.2 Subsidy worked examples and state support

Micron, Sanand (₹22,516 crore, about US$2.75 billion): centre 50% × 22,516 = ₹11,258 crore; Gujarat 40% × 11,258 = ₹4,503 crore (20% of cost); Micron 22,516 − 11,258 − 4,503 = ₹6,755 crore (30%), which matches Micron's stated US$825 million.

Tata–PSMC fab, Dholera (₹91,526 crore): centre 50% × 91,526 = ₹45,763 crore; Gujarat 40% × 45,763 = ₹18,305 crore; Tata and partner ₹27,458 crore (30%).

State support and talent. Gujarat adds a ₹2 per unit power subsidy for 10 years, water at ₹12 per m^3 for the first five years, a 50% subsidy on captive desalination plants and up to 75% land subsidy at Dholera. A 200 MLD seawater desalination plant for Dholera is at the DPR stage. Assam and Odisha also top up central support (40% and 50%). On talent, 315 institutions have EDA tools and about 67,000–70,000 students have been trained, against a target of 85,000.

Semicon 2.0's new areas. Materials, test facilities and equipment R&D get 30% capital support, with low entry thresholds (₹50 crore for materials); equipment makers get a 10% → 2% production incentive on domestically sourced content from 2028-29; design gets seed funding up to ₹15 crore, government equity co-investment

alongside VCs, royalty-based financing and a 9% deployment incentive (up to ₹30 crore per product); R&D and talent get up to 75% of project cost, including state support.

Targets. Neither the Cabinet note nor the gazette sets a technology-node or market-size target. The targets often quoted (3 nm by 2032, 2 nm by 2035, 70–75% self-sufficiency by 2029, a US$200 billion market by 2035) come from ministerial speeches, NITI Aayog's roadmap and industry reports, not from the scheme documents.

B.3 Malaysia and Vietnam

Malaysia has done OSAT since Intel arrived in 1972 and holds about 7% of global ATP capacity. Its National Semiconductor Strategy (2024) targets RM500 billion of investment and 60,000 engineers by 2030, backed by at least RM25 billion (about US$5.3 billion). Unusually, the strategy itself admits Malaysia is "over-concentrated" in OSAT and wants to move into design and advanced packaging. It also has local suppliers of test and automation equipment, such as ViTrox and Pentamaster (Ref. 14).

Vietnam is at a similar stage to India but further ahead. Intel's largest assembly-test plant is there, and Amkor has built a US$1.6 billion plant near Hanoi. Decision 1018 (2024) targets 100 design firms, one small fab, 10 packaging plants and 50,000 engineers by 2030 (secondary summaries). SIA–BCG expects Vietnam's ATP share to rise from under 1% to about 8% by 2032.

B.4 Support models and US CHIPS grants

Countries support chipmaking in different ways. Taiwan and Korea mainly use tax credits (Taiwan: 25% of advanced R&D and 5% of advanced equipment; Korea: 20% of facility investment for large firms). China uses state equity funds (Big Fund III, about US$47.5 billion). Malaysia uses capital grants worth about RM25 billion. India pays a direct cash grant during construction.

Exhibit A6: US CHIPS grants cover only about 9–20% of the investment they support

Company	Grant (US$B)	Investment supported (US$B)	Grant share
TSMC	6.6	65	10%
Intel	7.9	~90	9%
Micron	6.2	~50 (to 2030)	12%
Samsung	4.7	37	13%
Amkor (packaging)	0.4	~2	20%

Source: US Department of Commerce award releases; GAO-26-107882, Jul 2025 (Ref. 13): median 14% across 40 projects. With the 25–35% investment tax credit, total federal support is at most about 35–55% of capex.

B.5 The 12 approved units

Who makes what, for whom. CG Semi (Sanand) is a JV with Renesas, which takes close to half of capacity. Kaynes (Sanand) makes power modules for Alpha & Omega Semiconductor. Tata Assam (Jagiroad) is designed for 48 million chips a day, with Qualcomm and Nexperia lined up. HCL–Foxconn (Jewar) will package display-driver chips for phone and TV assembly. The smaller 2025–26 units cover SiC power, glass substrates (3D Glass Solutions), discretes (CDIL, Suchi), phone packages (ASIP) and GaN displays (Crystal Matrix).

Exhibit A7: The 12 approved units: ₹1,64,293 crore, three in commercial production

#	Company	State	Type	Capacity (as stated)	₹ crore	Approved	Status (Nov 2026)
1	Micron Technology	Gujarat	ATMP (memory)	1,352M units/yr (PRS); PIB ~14M/week	22,516	Jun 2023	Commercial production; captive
2	Tata Electronics + PSMC	Gujarat	Silicon fab	50,000 wafer starts/month	91,526	Feb 2024	Under construction; Nexperia MOSFETs planned
3	Tata Semiconductor Assembly and Test	Assam	OSAT	48.3M chips/day	27,120	Feb 2024	Pilot production; Qualcomm, Nexperia, ROHM

#	Company	State	Type	Capacity (as stated)	₹ crore	Approved	Status (Nov 2026)
4	CG Semi (CG Power + Renesas + Stars)	Gujarat	OSAT	15M chips/day	7,584	Feb 2024	Commercial production; Renesas ~50%
5	Kaynes Semicon	Gujarat	OSAT	6.3M chips/day	3,307	Sep 2024	Commercial production; Alpha & Omega
6	India Chip (HCL + Foxconn)	Uttar Pradesh	OSAT (display drivers)	20,000 wafers/month; 36M units/month	3,706	May 2025	Under construction; due 2028
7	SiCSem + Clas-SiC	Odisha	Compound fab + ATMP	60,000 wafers/yr; 96M units/yr	2,067	Aug 2025	Start not confirmed
8	3D Glass Solutions	Odisha	Advanced packaging	69,600 panels; 50M units; 13,200 3DHI modules/yr	1,944	Aug 2025	Start not confirmed
9	Continental Device India (CDIL)	Punjab	Discrete fab/ATMP	158M units/yr	118	Aug 2025	Start not confirmed
10	ASIP Technologies + APACT	Andhra Pradesh	OSAT	96M units/yr	469	Aug 2025	Start not confirmed
11	Crystal Matrix	Gujarat	Compound fab + ATMP	72,000 m² panels + 24,000 epi wafer sets/yr	3,068	May 2026	Start not confirmed
12	Suchi Semicon	Gujarat	OSAT (discretes)	1,033.2M chips/yr	868	May 2026	Pilot line since 2024
	Total				**1,64,293**		Tata 72%; Tata + Micron 86%; Gujarat 78%

Source: PRS Table 12; PIB and Cabinet briefings (Refs. 5–9); company releases. Workbook, Company and investment tracker tab (totals and shares by SUMIFS).

Annex C. Company case detail

C.1 Micron Sanand

Why India, in full.

1. **China risk (strongest).** On 21 May 2023 China's cyber regulator told critical infrastructure operators to stop buying Micron products. On 16 June Micron said a "low-double-digit percentage" of its worldwide revenue was at risk. Six days later, on 22 June, it announced India.[10] China-headquartered customers fell from about a quarter of Micron's revenue before the ban to 11% in FY2026. Micron did not leave China, though: in the same month it announced about US$600 million more for its Xi'an back-end plant. India is diversification, not an exit.
2. **Subsidy (strong).** The centre and Gujarat pay 70% of the project; Micron's share is up to US$825 million (Section 4.2).
3. **Back-end capacity for the second half of the decade (medium).** Micron said the plant was "to address demand for the latter half of this decade." Its new back-end capacity in Taiwan and Singapore is going into high-bandwidth memory (HBM) for AI, so conventional assembly needed a new home.
4. **Cost and talent (medium).** A factory worker in India costs about US$294 a month, against US$403 in Malaysia and US$510 in China (JETRO). Micron already has design centres in Bengaluru and Hyderabad.
5. **India's own market (weakest).** The first shipment went to Dell for laptops made in India. But Micron says the site serves customers worldwide, and its CEO said in September 2026 that Sanand's capacity already exceeds India's memory needs.

[10] Micron 8-Ks of 22 May and 16 Jun 2023 (Ref. 15).

Exhibit A8: Sanand went from approval to commercial production in under three years

Date	Milestone	Source
Jun 2023	Approval; MoU signed	Micron, PIB
Sep 2023	Ground broken	PIB (Feb 2026)
Feb 2024	Machine installation in pilot facility	PIB (Feb 2026)
Dec 2025	Pilot production	Micron FQ1-26 call
Feb 2026	Commercial production; first modules to Dell	Micron
2026 / 2027	"Tens of millions" of chips / "hundreds of millions"	Micron

Source: as listed; PIB release of 28 Feb 2026 (Ref. 8).

Is Sanand "10% of Micron's output"? Business Standard (secondary, paywalled) reported that Sanand at full capacity would be about 10% of Micron's "global output" (Ref. 34). The claim is plausible. In units: in a normal year (FY2025, US$37.4 billion) Micron sells roughly 5–7.5 billion chips at US$5–7 each, and Phase 1's ~728 million units would be 10–14% of that. In value: my base-case 2030 throughput (US$4.7 billion) equals about 12% of FY2025 revenue, but only 3.5% of boom-year FY2026 revenue. So "10%" depends on which year and which measure; Micron itself has never said it.

Memory cycle. Memory is the most cyclical part of the industry. Micron's gross margin fell from about 47% in 2021–22 to −33% in early 2023, then rose to 87% in the last quarter of FY2026. FY2026 revenue was US$133.2 billion, more than three times FY2025. The cause is AI: data-centre demand for HBM and server memory has outrun supply, and TrendForce's outlooks imply DRAM contract prices about six times their late-2025 level by the end of 2026. Over the last three cycles, margins took about six quarters on average to go from peak to trough; FY2026 Q4 is not yet confirmed as the peak.

Subsidy per job and per chip. On costs, the subsidy works out to about ₹3.2 crore per planned direct job (5,000 jobs; about 1,300 people at opening), and about 65% of it had not yet been recognised by Micron as of September 2026. Depreciation is small per chip: about US$0.29 per unit at full capacity without support (US$2.75 billion over 7 years), and US$0.09 with 70% support, against a unit value of about US$6.3. So the subsidy mattered to Micron as a saving on the capital it had to put in, not because it changes the cost of a memory chip. What India does not get from Sanand is a fab, memory IP or control over where the output goes.

Exhibit A9: Sanand's main risks come from its parent: the memory cycle, wafer supply and Micron's allocation decisions

Risk	Why it matters
Memory down-cycle	In 2023 margins went below zero; volumes and hiring would slow
One customer	Every unit goes to Micron, so Micron's allocation decides utilisation
Wafer supply	Output is limited by Micron's front-end growth
Geopolitics	Rests on US–India ties; it could shift if US policy pulls packaging onshore
Utilities, logistics	24x7 power and water; wafers flown in and products flown out
Talent	Fast ramp; about half of the ~1,300 staff at opening were fresh graduates
Subsidy timing	About 65% of government support not yet recognised (10-K); delays hit Micron's cash, not the plant's cost

Source: Micron FY2026 10-K (Ref. 15); company releases; news reports as cited in the workbook.

C.2 Tata Electronics

Exhibit A10: Dholera is a 50,000-wafer-a-month, 28–110 nm fab due for commissioning in 2028

Item	Detail	Source
Capacity	50,000 wafer starts per month, 300 mm	PIB (FSA, Mar 2025)
Nodes	28, 40, 55, 90, 110 nm (licensed from PSMC)	Tata release (Feb 2024)
Products	Power management ICs, display drivers, microcontrollers, logic; Nexperia MOSFETs planned	Tata and Nexperia releases
Cost	₹91,526 crore (~US$9.6 billion); 50% central support, pari-passu	PIB (FSA)
Timeline	Approved Feb 2024 → foundation Mar 2024 → PSMC agreement Sep 2024 → FSA Mar 2025 → ASML lithography MoU May 2026 → first fab commissioning **2028**	PIB (official, Jul 2026)

Source: as listed (Refs. 1, 5, 16).

Exhibit A11: Dholera's illustrative output value is about US$0.3 billion in 2030 and US$0.8 billion in 2032 in the middle case

Year	Wafers out (P10–P50–P90)	Illustrative output value at benchmark prices
2030	96k – 149k – 216k	US$0.19 – 0.32 – 0.50 billion (₹1,770 – 3,027 – 4,787 crore)
2032	295k – 380k – 459k	US$0.54 – 0.81 – 1.13 billion
Full capacity	570k (at 95% yield)	~US$1.26 billion at median price

Source: own Monte Carlo, 10,000 runs, inputs drawn independently (phase6_tata.ipynb). Not a Tata revenue forecast.

Dholera timing. Tom's Hardware (May 2026) reported civil work at about 50% and "trial production" still targeted for late 2026, despite a foundation redesign for soft, saline soil. The two dates are not really in conflict: trial production means running test wafers through the first installed tools; commissioning means qualified, saleable output. The ASML tool agreement was signed only in May 2026, so volume output before 2028 is unlikely. I use PIB's 2028 date.

Assam detail. The Jagiroad plant (₹27,000 crore, ~US$2.8 billion) is designed for 48 million chips a day using wire bond, flip chip and integrated system-in-package. Tata's original target was first-phase operations by mid-2025; the Assam CM said pilot chips started in October 2026, with an inauguration expected in December 2026–January 2027 (a weak source), roughly 16–19 months late. Named customers are Qualcomm automotive modules, ROHM's automotive power MOSFET and Nexperia discretes, with exploratory MoUs with Bosch, Intel and L&T Semiconductor. Assam will reach volume two years before Dholera. It is Tata's first real test of execution and customer qualification.

PSMC detail. Under the September 2024 agreement PSMC provides fab design and construction support, a licence for its process technologies and engineering support to transfer them. It lost money in 2023–25 (net loss NT$7.8 billion in 2025), with a 2025 gross margin of −3% and utilisation of 71–82% through 2024–25. By 3Q26 it ran at 97% utilisation with a 48% gross margin. Its nine-month 2026 profit also includes NT$17.2 billion of other operating income, the period in which it divested its P5 fab (Refs. 17, 31).

Capex benchmark. Dholera costs about US$193k per wafer-per-month of capacity. UMC's Singapore Fab 12i P3 (22/28 nm) was announced at US$5 billion for 30,000 wspm, about US$167k; Hua Hong's new Wuxi fab costs about US$81k, but sits in a low-cost Chinese cluster. So Dholera is about 16% above a comparable non-Chinese fab. Part of that is scope (land, utilities, a greenfield site); part may be a first-fab premium.

China price risk, both ways. In 2024–25 mature-node prices were weak: customers held excess inventory under long-term contracts, and Chinese foundries, backed by the state, cut prices hard. In 2026 the market tightened: TSMC and Samsung are cutting 8-inch and some 12-inch mature capacity, and AI servers need many more power ICs. Top-10 8-inch utilisation is about 88–90%, prices rose 5–15% in the first half of 2026 with a third round planned into 2027, and PSMC has 80% of its 2027 MOSFET/IGBT capacity under long-term contracts. The risk for Dholera is that nearly 70% of new 12-inch mature-node capacity is being built in China (Ref. 31).

Exhibit A12: Wafers, tools and most materials stay imported; only Fujifilm has committed money to a local plant

Input	Source today	Localisation steps (MoUs, mostly "explore")
Lithography and process tools	Imported (ASML, TEL)	Service, spares and training in India
Tool hook-up	Kelington (Malaysia)	Local engineering talent
Silicon wafers	Imported	None announced
Photoresists, high-purity chemicals	Imported	Fujifilm ₹800 crore (US$83M) materials plant in Dholera; Sumitomo to explore local manufacture
Specialty gases, chemical delivery	Imported	Merck: warehousing and supply-chain localisation
Chemical logistics	–	Sojitz and NRS (storage and transport framework)
Design enablement (PDKs, TCAD)	Synopsys	–
Packaging materials (Assam)	Imported	Enomoto to explore lead frames in India

Source: Tata Electronics releases 2024–26 (Ref. 16). Also in the workbook, Company and investment tracker tab.

Exhibit A13: Customer acquisition is Dholera's biggest risk, ahead of Chinese pricing and PSMC dependence

Risk	Likelihood	Impact	Why
Customer acquisition / low utilisation	High	High	Biggest driver of output; global buyers choose foundries; slow qualification
Mature-node price pressure from China	Medium	High	Prices rising in 2026 on AI power demand, but China leads 12-inch mature-node expansion; a 2028–30 down-cycle would hit as Dholera ramps
Yield learning curve	Medium	Medium	No Indian fab experience; first-year yields drive 2028–29 output
Technology transfer / PSMC dependence	Medium	High	Single licensor; PSMC is mid-sized and focused on its own AI products
Execution delay	Medium	Medium	Soil redesign already needed; Assam ~16–19 months late
Talent	Medium	Medium	Fab process engineers are scarce in India; early dependence on PSMC's engineering support
Subsidy disbursement timing	Low–Medium	Medium	ISM spent only ~7% of the Semicon 1.0 outlay by FY26, but payments are tied to construction

Source: own assessment based on the evidence in Section 8.

Annex D. Key assumptions

Exhibit A14: Key assumptions behind the model (all on the workbook's Inputs tab)

Assumption	Value (low / base / high)	Type	Basis
Exchange rate	₹95 per US$	Project rule	One rate; company US$ figures kept in US$
India chip market growth, 2025–30	9% / 13% / 18% a year	Assumed	Global normal growth; IESA; Deloitte
Sub-assembly imports and chip content	US$30–40B; 35–45%	Assumed	Own assumption, no source (Annex A.2)
OSAT revenue ÷ capex at full ramp	0.625	Calculated	CG Semi US$500M target ÷ ₹7,600 crore at ₹95
Merchant OSATs' share of full ramp in 2030	40% / 60% / 80%	Assumed	Start dates 2026–28
Share of output sold in India	Micron 10 / 20 / 30%; others 20 / 30 / 40%	Assumed	Micron says the site serves the world
Sanand utilisation in 2030 (of 1,352M units)	35% / 55% / 80%	Assumed	55% ≈ Phase 1 running full
Sanand value per unit	DRAM US$3–5; NAND US$6–12; SSD US$40–80 (hand-built)	Assumed	TrendForce pre-boom contract prices
Fab price per 12-inch-equivalent wafer	US$992 / 2,208 / 3,343	Calculated	Peer min / median / max, 2021–25
Dholera utilisation of design capacity, 2030	10% / 25% / 50%	Assumed	Commissioning 2028
Mature-node share of India's demand	30% (25–35%)	Assumed	WSTS 2025 product split; no India split exists
Fab fixed share of manufacturing cost	63.9%	Actual	UMC 20-F FY2023
Fab R&D + SG&A	11.3% of revenue	Actual (likely low)	UMC 2025
Effective Semicon 1.0 + Gujarat support	~65% in present value	Calculated	Gujarat paid in five instalments, 10% discount
Semicon 2.0 fab support with Gujarat	56% of eligible capex (~45% of total)	Calculated	Land, IDC and licence not eligible
Hurdle rate for comparison	10–12%	Assumed	My assumption, not sourced
Global ATP market growth to 2030	7.3% a year from US$95B	Assumed	Normal-cycle global growth

Source: workbook, Inputs tab, where each line carries its source and type.

Annex E. Data cautions and limitations

Data cautions

- **Sanand capacity.** PRS gives 1,352 million units a year; PIB gives about 14 million a week (~728 million a year), probably Phase 1 only. Unresolved.
- **WSTS.** The Autumn 2026 forecast is not yet released; global figures use the Spring 2026 forecast and the Q2 update (6 Aug 2026).
- **Global capacity data.** India is not split out in SIA–BCG or SEMI data. SIA–BCG 2024 (2022 data) is the latest regional split found. Global unit counts (TechInsights O-S-D plus a 32% IC share) should be treated as ±30%.
- **Missing official files.** Several PIB approval notes were not available; tracker rows use PRS Table 12, Cabinet briefings via news, and company releases, each labelled in the workbook.

- **Secondary sources.** Vietnam's Decision 1018 is from English summaries; Korea's package is from news; Tata group financials (other than the Tata Sons annual report and AOC-1) are from ET via Angel One; the Assam start date is from a weak source.
- **Peer data.** Tearsheet and company figures differ for PSMC, Hua Hong and VIS gross profit; company figures are used. Some utilisation figures are derived. GlobalFoundries and PSMC revenue includes some non-wafer income, so their revenue per wafer is overstated.
- **Micron quarterly data.** FY2025–26 quarterly rows are from FMP via Bigdata.com and rebuild the 10-K totals. Call-mention counts depend on the keyword index.
- **Prices.** All Sanand value-per-unit figures are assumed, anchored to TrendForce price lists; they are not Micron prices.

Limitations

- Both reference facilities are built from peer cost structures, not Indian plant data. Subsidy cases ignore tax, working capital and financing costs.
- The OSAT model assumes CG-level revenue per rupee of capex for all merchant OSATs. Kaynes, HCL–Foxconn and the compound-semiconductor units may differ. OSAT support cases are upper bounds because CG's land and IDC are not broken out.
- Share sold in India and value-added shares are my own assumptions. They are the weakest inputs in Section 9.1, which is why I show all three measures.
- UMC's 11.3% R&D and SG&A is likely low for a new entrant, so fab IRRs are flattered.
- The fab IRR excludes any value from the ecosystem it creates (suppliers, talent, strategic security). This is the main argument for funding it, and it cannot be captured in a project IRR.
- The 30% mature-node share of India's demand is an assumption; no source splits India's demand by node.

Annex F. Glossary

Term	Meaning
ATMP / OSAT	Assembly, testing, marking and packaging / outsourced semiconductor assembly and test: the back-end step that turns a finished wafer into packaged chips
ATP	Assembly, test and packaging (SIA–BCG's term for the same step)
Fab / front end	The plant that makes chips on silicon wafers
Fabless, foundry, IDM	A firm that only designs chips; a firm that only makes chips for others; a firm that does both
Node (nm)	The generation of manufacturing technology; smaller numbers are more advanced. Below about 7 nm needs EUV lithography
Mature node	28 nm and above; used for power, analog, display-driver and microcontroller chips
wspm	Wafer starts per month, the usual measure of fab capacity
12-inch equivalent	One 12-inch (300 mm) wafer = 2.25 eight-inch wafers by area
Utilisation / loading	Output as a share of capacity; "loading of installed capacity" counts only tools already installed
Yield	Good wafers or chips as a share of those started
Break-even loading	The loading at which gross margin is zero
Pari-passu	Government support released in step with the company's own spending
Eligible capex	The part of project cost a scheme will support; under Semicon 2.0 it excludes land, IDC and technology transfer
IDC	Interest during construction
DLI	Design Linked Incentive scheme
IRR	Internal rate of return: the discount rate at which a project's cash flows sum to zero
P10 / P50 / P90	The values below which 10%, 50% and 90% of simulation runs fall
ρ (rho)	The correlation between two simulated inputs
Throughput value	The value of products passing through a captive plant at market prices; not the plant's revenue
Illustrative output value	Dholera's output valued at peer benchmark wafer prices; not a Tata revenue forecast

Annex G. Notebooks and workbook

Exhibit A15: Every number in the report can be reproduced from one workbook and five notebooks

File	What it does	Sections
India_Semiconductor_Model.xlsx	Inputs, market sizing, tracker, benchmarking, financial scenarios, sources; live formulas, scenario and support selectors	3–9
phase2_trade.ipynb	TradeStat import series, solar adjustment and gap table	3
phase4_tracker_sql.ipynb	The 12-unit tracker in SQLite with concentration queries	6
phase5_micron.ipynb	Micron cycle statistics, call mentions, Sanand scenarios and Monte Carlo	7
phase6_tata.ipynb	Foundry peer dataset, break-even estimates, Dholera ramp Monte Carlo	8
phase7_models.ipynb	Reference OSAT and fab models, correlated Monte Carlo, India 2030 local supply	9
chart_sanand_mc_corr.py	Exhibit 10 (Sanand Monte Carlo at ρ = 0 and 0.7)	7

All notebooks use pandas, numpy and matplotlib only, and each dataset is a CSV with a source column. They will be collected in one GitHub repository with a README that links each chart to its exhibit.